Beyond Self-Consistency: Self-Extracted Constraints Provide No Detectable Improvement in LLM Reasoning

Anonymous ACL submission

Can symbolic verification via a provably sound solver improve Large Language Model (LLM) reasoning when constraints are self-extracted from the model's own traces? Across six confirmatory conditions ($n \geq 200$), we find no detectable improvement exceeding $\sim$2 percentage points (pp). We introduce Symbolic Invariant Consensus Aggregation (SICA), which extracts logical constraints from chain-of-thought traces via weighted MAX-SAT. Despite Z3 validating 99% of self-extracted constraints, SICA matches majority voting on over 90% of problems across six models (7B–32B) and four benchmarks. No condition shows a significant positive effect after FDR control; four of six powered conditions confirm equivalence within ± 2 pp (TOST); the remainder are underpowered for smaller effects. The failure is a robust empirical regularity: self-extracted constraints track the majority answer—a property we term *symbolic confirmation bias* (bias ratios 1.66–$4.97\times$ across nine models, 7B to frontier). Under source-attribution scoring and degeneration, SICA reduces to majority vote (Observation 1); the $M \to T \to A \to C$ Markov chain bounds self-extracted information, revealing a deeper mechanism underlying Huang et al. (2024)'s findings. Twenty-seven training-free variants fail due to a persistent independence-accuracy trade-off. As exploratory evidence, cross-architecture verification escapes the bias ($+6.67$ pp on the weakest generator, $p < 0.001$). Three zero-cost diagnostics characterize failure modes across all tested conditions.

1 Introduction

Self-Consistency (SC) (Wang et al., 2023) selects the majority answer across K chain-of-thought traces, a strategy grounded in the Condorcet Jury Theorem: independent voters converge to truth (de Condorcet, 1785; Ladha, 1992). In practice, SC's gains plateau for strong models (Loo et al., 2025), pointing to correlated traces that reduce the effective ensemble size. A natural remedy is to move beyond simple vote-counting: extract logical constraints from the traces and let a provably sound solver adjudicate. If the solver can filter contradictions and enforce logical consistency, it should recover information that raw majority voting discards.

We test this intuition through Symbolic Invariant Consensus Aggregation (SICA), which extracts logical constraints from reasoning traces (prompted as first-order logic; 99% are propositional at the Z3 level) and selects answers via weighted partial MAX-SAT (§3.1; Figure 1). Despite Z3 validating nearly all self-extracted constraints, SICA produces answers identical to majority voting on the vast majority of problems. Across six models (7B–32B), four benchmarks, and multi-seed replication, no condition shows a significant positive effect after FDR control (Table 1). Four of six powered conditions confirm equivalence within ± 2 pp (TOST); the remainder are underpowered for smaller effects.

The failure is not in the solver but in the constraints it operates on. Self-extracted constraints systematically support the model's majority answer—a robust empirical regularity of zero-shot self-extraction pipelines we term *symbolic confirmation bias* (§5.1). Because each trace determines both an answer and its constraints, the formal channel reduces to a weighted echo of the majority vote. We show this degeneration is robust (Observation 1) and confirm it holds across nine models from 7B to frontier scale. Prior work shows LLMs cannot self-correct reasoning without external feedback (Huang et al., 2024); our analysis reveals a deeper mechanism underlying these findings, providing a Markov chain formalization and actionable diagnostics for self-verification.

Twenty-seven training-free variants within the constraint-extraction paradigm confirm the trade-off is fundamental (§5.2): methods that break

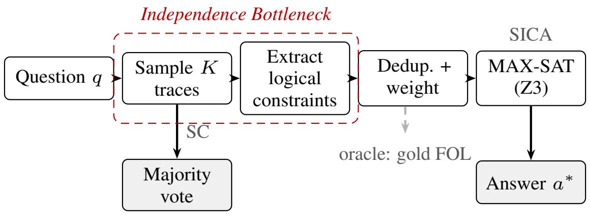

Figure 1: The SICA pipeline. SC selects the majority answer from K traces. SICA extracts logical constraints, deduplicates, and solves MAX-SAT via Z3. Red dashed box: the independence bottleneck. Dashed arrow: oracle variant with gold FOL.

trace correlation destroy accuracy, while methods that preserve accuracy fail to achieve independence. Cross-architecture verification provides preliminary evidence of escape: a structurally independent verifier yields significant gains when it exceeds the generator's accuracy, with benefit governed by a capability-gap pattern between verifier and generator (§6). Three zero-cost diagnostics enable practitioners to predict a priori whether any self-extraction-based aggregation method will help (§5.3).

Our contributions are:

1. **The symbolic mirage.** Provably sound Z3 verification becomes vacuous under self-extraction. We diagnose the underlying mechanism: self-extracted constraints inherit the generating model's distributional prior (symbolic confirmation bias), reducing SICA to majority vote under source-attribution scoring (Observation 1). The pattern holds across nine models and 27 training-free variants (§5.1, §5.2).[1]
2. **Diagnostic framework.** Three zero-cost diagnostics characterize failure modes across all tested conditions within the self-extraction paradigm, with prospective validation on held-out model families, providing practitioners with a priori failure-mode diagnosis from a pilot run (§5.3).
3. **Cross-architecture escape.** Preliminary evidence that structurally independent verification can break the bottleneck, with gain governed by a capability-gap pattern between verifier and generator (§6).

2 Related Work

Self-consistency and ensemble theory. Self-consistency (SC) selects the most frequent answer across chain-of-thought traces (Wang et al., 2023). Extensions address free-form generation (Chen et al., 2024), confidence weighting (Taubenfeld et al., 2025), diverse prompting (Li et al., 2023), and reasoning-aware filtering (Wan et al., 2024; Toh et al., 2024; Kim et al., 2026). SC gains plateau early (Loo et al., 2025; Byerly and Khashabi, 2025). The Condorcet Jury Theorem (de Condorcet, 1785) guarantees majority-vote convergence under independence; ensemble theory (Dietterich, 2000; Krogh and Vedelsby, 1994) establishes that ensemble gain requires error diversity. Ladha (1992) show correlated voters reduce effective ensemble size; our K_{eff} adapts Fleiss' κ from diversity metrics (Kuncheva and Whitaker, 2003; Brown et al., 2005) to quantify this compression (§3.1). Kim et al. (2025) demonstrate correlated errors in LLM outputs; Kuai et al. (2026) propose cross-model independence auditing, complementary to our within-model process-κ.

Neuro-symbolic reasoning. We distinguish three paradigms: *specification-generation* (Pan et al., 2023; Olausson et al., 2023; Ye et al., 2024) and *program-generation* (Gao et al., 2023; Chen et al., 2023b; Lyu et al., 2023) formalize *before* answer generation; *trace-extraction* (this work) operates post-hoc, inheriting confirmation bias (§5.1). Recent extensions include symbolic planning and verified reasoning (Ryu et al., 2025; Xu et al., 2026; Li et al., 2025; Wang et al., 2025; Singh et al., 2026; Luo et al., 2025; Miya, 2025); Amonkar et al. (2025) show formalization quality degrades with problem complexity; Chan et al. (2026) argue logical soundness alone is unreliable. Prior studies note high formalization error rates but treat them as quality problems; we show the errors are *systematically biased* toward the model's answer (Observation 1; Figure 3). Our analysis reveals a deeper mechanism underlying Huang et al. (2024)'s finding that LLMs cannot self-correct without external feedback, showing that even Z3-validated constraints carry no information beyond majority voting.

Distinction from CoT confirmation bias. Symbolic confirmation bias (SCB) is complementary to, but structurally distinct from, CoT-level confirmation bias (Wan et al., 2025; Turpin et al., 2023; Jhaveri et al., 2026). CoT bias operates at the *trace level*—individual chains may not faithfully reflect the model's computation—while SCB operates at the *aggregation level*: even if each trace is locally faithful, the *collection* of self-extracted constraints mirrors the answer distribution.

[1] Code and data will be released upon acceptance.

Relationship to Huang et al. (2024). Huang et al. empirically demonstrate that LLMs cannot self-correct reasoning without external feedback; our work reveals a deeper mechanism underlying this finding. Where they report the phenomenon through evaluation of prompting strategies, we formalize the failure via the $M \to T \to A \to C$ Markov chain (Observation 1) that explains *why* self-correction fails. Where they lack tools for a priori prediction, we provide three zero-cost diagnostics (K_{eff}, BR, C_0) that predict failure modes before deployment. Where they do not test formal verification, we systematically evaluate 27 training-free variants including Z3-validated constraints, showing that even provably sound solvers cannot escape the bottleneck.

Test-time compute and self-verification. Scaling test-time compute via additional traces (Snell et al., 2025), structured search (Yao et al., 2023), or extended reasoning (DeepSeek-R1 (Guo et al., 2025), QwQ (Qwen Team, 2024), OpenAI o1/o3) faces the same correlation challenge when traces share a single model's biases. Multi-agent debate (Liang et al., 2023; Du et al., 2024), self-evaluation and refinement (Xie et al., 2023; Weng et al., 2023; Madaan et al., 2023; Dhuliawala et al., 2023), iterative self-improvement (Shinn et al., 2023; Zhou et al., 2024), tool-augmented critique (Gou et al., 2024), formal CoT verification (Zhu et al., 2025; Chen et al., 2025), and multi-agent verification (Lifshitz et al., 2025) report gains, but none tests self-extracted constraints under formal solvers (cf. Kamoi et al., 2024). We find the bottleneck lies in constraint extraction, not verification: self-extracted constraints inherit the model's bias regardless of solver soundness (§5.1). Oracle controls confirm the verification mechanism itself is sound (Appendix G).

Training-based verification. Trained verifiers escape the self-extraction bottleneck via external reward signals: outcome and process reward models (Cobbe et al., 2021; Lightman et al., 2024; Wang et al., 2024), discriminator-guided decoding (Khalifa et al., 2023), RL-based self-verification (Zeng et al., 2025), dual preference optimization (She et al., 2025), and execution-based filtering (Li et al., 2022; Chen et al., 2023a). These operate outside the pipeline of Observation 1, potentially breaking the DPI bound by violating the trace-determined assumption. The failure modes differ: SICA's bottleneck is extraction fidelity (constraints echo the model's prior), whereas PRM/ORM's is reward model generalization (Wang et al., 2024). Our diagnostics (BR, K_{eff}) may extend to learned verification by quantifying whether learned scores introduce genuine independence.

3 Method

3.1 SICA Pipeline

Self-Consistency (SC) (Wang et al., 2023) samples K reasoning traces and selects the majority answer. Under correlated traces (Fleiss' κ (Fleiss, 1971)), effective sample size reduces to

$$K_{\text{eff}} = K \,/\, [1 + (K - 1) \cdot \kappa] \qquad (1)$$

(following Ladha (1992)). Intuitively, K_{eff} measures how many effectively independent votes K correlated traces contribute; e.g., $K{=}12$ traces with $\kappa{=}0.69$ yield $K_{\text{eff}} \approx 1.3$, meaning twelve correlated traces carry no more independent information than 1.3 uncorrelated ones. This motivates constraint-based aggregation.

Symbolic Invariant Consensus Aggregation (SICA) extends SC by extracting formal constraints from traces and selecting the answer best supported by the satisfiable subset (Figure 1).

Trace generation. Given question q, an LLM generates K traces $\{t_1, \ldots, t_K\}$ at temperature $T = 0.7$. Each trace t_i yields answer a_i.

Constraint extraction. The same LLM extracts logical constraints from each trace at $T_{\text{ext}} = 0.3$. The extraction prompt requests JSON output with constraint type, expression, and Z3-compatible formula (Appendix A); in practice, 99.0% of extracted constraints are propositional at the Z3 level, with only 1.0% using true quantifiers. The *extraction rate* is the fraction of traces yielding at least one parseable constraint.

Deduplication and MAX-SAT. Constraints are deduplicated by syntactic equivalence (global average deduplication ratio: 46.0%); each unique constraint c_m receives weight w_m equal to the number of source traces. The objective finds the largest-weight satisfiable subset:

$$\mathcal{C}^* = \arg\max_{\mathcal{S} \subseteq \mathcal{C}} \sum_{c \in \mathcal{S}} w_c \quad \text{s.t.} \quad \mathcal{S} \text{ is satisfiable.} \qquad (2)$$

Z3 (de Moura and Bjørner, 2008) solves this with a 10-second timeout. Answer assertion constraints (`answer_a == True`, weight 1) ensure every candidate receives a nonzero base score, making SICA a strict generalization of SC.

Scoring. Let $\mathcal{C}^{\setminus} = \mathcal{C} \setminus \mathcal{C}^*$. Each candidate a is scored by:

$$\text{score}(a) = \sum_{c \in \mathcal{C}^*} w_c \cdot \sigma(c, a) - \alpha \sum_{c \in \mathcal{C}^{\setminus}} w_c \cdot \sigma(c, a), \tag{3}$$

where $\sigma(c, a) = 1$ if c was extracted from any trace producing answer a, and $\alpha = 0.5$. The selected answer is $a^* = \arg\max_a \text{score}(a)$.

Degeneration condition. If self-extracted constraints are coupled to their source answer, MAX-SAT retains nearly all ($|\mathcal{C}^{\setminus}| \approx 0$), and scoring degenerates to:

$$\text{score}(a) \approx \sum_{\substack{c \in \mathcal{C}^* \\ \text{source}(c) \to a}} w_c, \tag{4}$$

equivalent to majority voting. We show in §5.1 that this degeneration occurs in practice.

The C_0 diagnostic. The multi-candidate rate $C_0 = |\{q : |\mathcal{A}(q)| > 1\}|/|\mathcal{Q}|$ is a zero-cost diagnostic: $C_0 < 5\%$ means SICA reduces to SC. $\alpha = 0.5$ is not critical (± 0.49 pp across $\alpha \in [0, 1]$).

Bias ratio. We quantify extraction bias via the *bias ratio*:

$$\text{BR}(M, T) = \frac{P(c \text{ supports } a_M \mid a_M \neq a^*)}{P(c \text{ supports } a^* \mid a_M \neq a^*)}, \tag{5}$$

where c is a self-extracted constraint, a_M the model's majority answer, and a^* the ground truth. BR quantifies the degree of extraction degeneration under source-attribution scoring (Observation 1): values above 1 reflect the trace-count base rate inherent to source-attributed constraints. BR $= 1$ indicates no coupling beyond chance. We validate this metric empirically in §5.1.

3.2 Experimental Setup

Models. Qwen2.5-14B-Instruct, Qwen2.5-7B-Instruct, Qwen3-14B (non-thinking mode) (Yang et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), LLaMA-3.1-8B-Instruct (Dubey et al., 2024), and QwQ-32B (Qwen Team, 2024) (7B–32B). We also evaluate the universality of symbolic confirmation bias on six additional models: gpt-4o uses self-extraction via API; four frontier models (gpt-4.1, gpt-5.5, o3, Gemini-2.5-Pro) use cross-model extraction where Mistral-7B extracts constraints from target traces; DeepSeek-R1-Distill-Qwen-8B likewise uses Mistral-7B cross-extraction (Table 2).

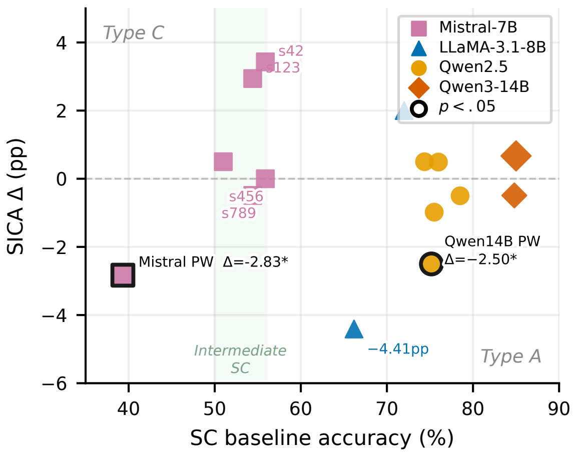

Figure 2: SICA Δ (pp) vs. SC baseline. Filled: $n \geq 100$; open: $n < 100$. The intermediate-SC region (50–60%, shaded) contains the only positive trends (none reaching $p < 0.05$). Type C: nominally significant degradation (raw $p = 0.014$; BH adj. $p = 0.042$).

Datasets. FOLIO (Han et al., 2022) (n=204, gold logical annotations), ProofWriter depth-5 (Tafjord et al., 2021) (n=600), LogiQA (Liu et al., 2020) (n=200), StrategyQA (Geva et al., 2021) (n=200). GSM8K (n=30) and AIME 2024 (n=22) probe the degenerate regime ($C_0 < 5\%$).

Parameters. K=12 traces at T=0.7, extraction at T_{ext}=0.3, α=0.5, Z3 timeout 10s. McNemar exact test (α=0.05); 95% bootstrap CIs (10,000 resamples) for $n \geq 200$. SICA incurs $4.5\times$ SC wall-clock; extraction accounts for 76% of overhead.

4 Results

Table 1 reports accuracy across all non-degenerate self-extraction conditions sorted by SC baseline. Two confirmatory conditions show significant degradation after BH FDR control (m=6; multi-seed aggregates and extensions are exploratory): Mistral-7B on ProofWriter (-2.83 pp, raw $p = 0.014$, adj. $p = 0.042$) and Qwen2.5-14B on ProofWriter (-2.50 pp, raw $p = 0.011$, adj. $p = 0.042$). No condition shows a significant positive effect. An aggregate sign test yields 8 positive and 7 negative Δ values ($p = 1.00$, two-sided). When SICA and SC disagree, SC selects the correct answer $2.6\times$ more often (139 vs. 53 across conditions), indicating that constraint-based reweighting introduces noise rather than corrective signal.

Figure 2 reveals regime structure. Fleiss' κ ranges from 0.21 to 0.82, yielding $K_{\text{eff}} = 1.20$–3.62 at $K = 12$ (Table 19, Appendix). Three regimes emerge: Type C (SC

Table 1: SICA vs. SC across all non-degenerate self-extraction conditions (sorted by SC). Two conditions show significant degradation after BH correction (m=6 confirmatory; remaining exploratory); no condition shows a significant gain. Δ = SICA − SC (pp). Symbols: ^{b}gold ICL; **bold**: $p < 0.05$; * BH adj. $p = 0.042$; ^{3s} 3-seed aggregate; ^{4s} 4-seed confirmatory aggregate (unmarked: single-seed exploratory; s123/s456/s789 are individual seed runs). All conditions have McNemar power $< 80\%$ for $|\Delta| = 2$ pp (Appendix D). Degenerate conditions ($C_0 < 5\%$: QwQ-32B AIME n=22, Qwen2.5-14B GSM8K n=30) yield $\Delta = 0$ and are reported separately in Appendix G.

Model	Domain	n	SC (%)	SICA (%)	Δ (pp)	p
Type C: error amplification (SC ≤ 40%)						
Mistral-7B	PW-D5	600	39.33	36.50	**−2.83**	0.014*
Intermediate SC (SC ∼50–60%, Mistral-7B only)						
Mistral-7B	LogiQA	200	50.00	51.50	+1.50	0.581
Mistral-7B	FOLIO (s123)	204	54.41	57.35	+2.94	0.070
Mistral-7B^{3s}	FOLIO	204	56.53	56.70	+0.16	1.000
Mistral-7B	FOLIO (s456)	204	55.88	55.88	0.00	1.000
Mistral-7B	FOLIO (s789)	204	59.31	56.86	−2.45	0.267
Type A: confirmation bias (SC ≥ 60%)						
LLaMA-3.1-8B	FOLIO	204	66.18	61.76	−4.41	0.093
LLaMA-3.1-8B	StrategyQA	200	72.00	74.00	+2.00	0.424
Qwen2.5-14B	FOLIO (ICL)^b	199	74.37	74.87	+0.50	1.000
Qwen2.5-14B	PW-D5	600	74.00	71.50	**−2.50**	0.011*
Qwen2.5-14B^{4s}	FOLIO	204	75.01	73.78	−1.24	1.000
Qwen2.5-7B	FOLIO	204	75.98	76.47	+0.49	1.000
Qwen2.5-14B	LogiQA	200	79.00	78.00	−1.00	0.625
Qwen3-14B	FOLIO	204	83.33	84.80	+1.47	1.000
Qwen3-14B	PW-D5	600	85.83	85.67	−0.17	1.000
Qwen3-14B (think)	FOLIO	204	86.27	86.76	+0.49	1.000

$\leq 40\%$), where low accuracy causes error amplification; an intermediate band (50–60%), containing the only positive trends; and Type A (SC $\geq 60\%$), where symbolic confirmation bias dominates. Seed-level noise is substantial: across three Mistral-7B seeds, Δ ranges from $+2.94$ to -2.45 pp despite $\kappa \in [0.45, 0.51]$. We trace the mechanism in §5.

TOST equivalence tests (Lakens, 2017) with pre-specified margin ± 2 pp confirm four conditions as equivalent; ten remain underpowered for effects below ~ 2 pp (MDE 1.86–4.84 pp; sensitivity analysis in Appendix D, U). Absence of evidence is not evidence of absence for small effects. Multi-seed replication confirms robustness: mean $\Delta = +0.16 \pm 2.70$ pp across three Mistral-7B seeds (Table 42); four Qwen2.5-14B seeds yield $\Delta \in [0.00, -2.01]$ pp. Oracle controls with gold annotations confirm the MAX-SAT mechanism is sound ($+29.90$/$+13.73$ pp FOLIO, $+17.17$ pp ProofWriter; $p < 0.001$), yet self-extraction captures only $\sim 12\%$ of this ceiling (Appendix G).

5 The Symbolic Mirage

5.1 Symbolic Confirmation Bias

Self-extracted constraints systematically align with the model's majority answer—a persistent empirical property of zero-shot self-extraction pipelines, not a cognitive bias analogy. On Mistral-7B FOLIO, when SICA errs, 83.3% of constraints support the incorrect answer—bias ratio $4.97\times$ (Table 16)—and 94% of errors overlap with SC. Z3 rejects only 3.4% of constraints, and MAX-SAT produces answers identical to count-only aggregation on $\geq 99\%$ of problems across all weight settings α (Table 11), confirming that the formal verification layer is vacuous (Eq. 4): 99% of constraints pass Z3 validation yet add no information beyond majority voting.

Crucially, BR > 1 is not an independent finding but a corollary of the degeneration identified in Observation 1: under source-attribution scoring, constraint weight is proportional to trace count, so the majority answer necessarily receives more support. A permutation null model—randomizing constraint–answer associations while preserving per-trace counts (1,000 permutations)—confirms this: $\mathrm{BR}_{\mathrm{null}} \approx \mathrm{BR}_{\mathrm{obs}}$ ($\mathrm{BR}_{\mathrm{excess}} \approx 0$; $p > 0.7$) across all tested conditions. BR thus quantifies the degree of extraction degeneration—how completely constraints collapse to majority-vote equivalents—not an independent bias signal. On problems where SC is correct, constraints still track the majority ($\mathrm{BR}_{\mathrm{correct}}$ 1.29–3.64$\times$), confirming that self-extracted constraints reflect distributional frequency rather than logical validity.

Table 2: Symbolic confirmation bias across nine models on FOLIO ($n = 204$). BR: mean per-problem bias ratio. n_{err}: number of incorrect instances on which BR is computed. Δ: SICA $-$ SC (pp). Ext.: [a]self-extraction; [b]cross-model (Mistral-7B extracts FOL from target traces). *Interim ($n = 95/204$). †8 API-failure questions excluded ($n = 196$). All models exhibit BR > 1; BR estimates for high-accuracy models ($n_{\text{err}} < 40$) have wider confidence intervals; Gemini ($n_{\text{err}} \approx 12$) should be interpreted with particular caution. Process-level similarity (κ_{proc}) ranges from 0.25 (R1-Distill) to 0.79 (gpt-4o/4.1/Gemini); see Table 21.

Model	Ext.	SC (%)	n_{err}	BR	Δ (pp)
Mistral-7B	a	55.88	90	4.97	0.00
LLaMA-8B	a	—	—	2.88	—
Qwen-14B	a	75.98	49	3.85	−0.49
R1-Distill-8B	b	74.51	52	2.92	+2.45
gpt-4o	a	76.96	47	2.65	−1.96
gpt-4.1	b	82.84	35	2.70	−1.47
gpt-5.5	b	84.31	32	3.48	+0.49
o3†	b	82.65	34	1.66	0.00
Gemini-2.5-Pro*	b	87.37	12	2.79	−1.05

The pattern generalizes across nine models from 7B to frontier (BR 1.66–4.97×; Table 2) and all eight cells of a 2×2 generator×extractor matrix (range 1.94–2.70; Appendix W). Constraints use trace-specific variables (56.4% zero variable overlap across answer groups), collapsing aggregation to majority voting.

Two controls rule out answer leakage: BR is uniform across reasoning steps (Fisher $p = 0.71$), and removing answer sentences before extraction yields no BR decrease (Wilcoxon $p \geq 0.12$) despite accuracy loss (McNemar $p < 0.013$). The bias reflects a distributional prior embedded throughout reasoning, not answer contamination.

Reasoning-process overlap exceeds answer-level correlation: mean process similarity (pairwise TF-IDF cosine) reaches 0.69 on Mistral-7B FOLIO, far exceeding answer-level κ (0.41–0.51). The gap generalizes across four families (Table 21). Under cross-model extraction, SICA gain decreases monotonically with process similarity ($\rho = -1.0$, $p = 0.008$; Table 17; conditional correction analysis in Appendix C).

These patterns—universal BR > 1, structural equivalence to trace-count distributions, and process-level correlation exceeding answer-level agreement—follow from a persistent empirical property of zero-shot self-extraction pipelines.

Observation 1 *(Argmax Preservation).* Let $t_1, \ldots, t_K \overset{\text{i.i.d.}}{\sim} P_M(\cdot \mid q)$ with trace-determined answers $a_i = h(t_i)$ and $n_a = |\{i : a_i = a\}|$. Under (i) the degeneration condition $|\mathcal{C}^{\setminus}| \approx 0$ (Eq. 4), (ii) source-attribution scoring where $\sigma(c, a)$=1 iff c originates from a trace producing a (Eq. 3), and (iii) approximately uniform constraint yield per trace within each answer group, SICA reduces to majority vote: $\arg\max_a \text{score}(a) = \arg\max_a n_a$.

Empirical support. The degeneration condition is quantified by the Z3 rejection rate: $|\mathcal{C}^{\setminus}|/|\mathcal{C}| \approx 0.034$, confirming $|\mathcal{C}^{\setminus}| \approx 0$. Source-attribution restricts each constraint's contribution to its originating answer. For condition (iii), per-trace constraint yield averages 5.6–6.0 constraints across answer groups (median within-problem CV=0.26; Appendix S), making total weight approximately proportional to n_a. SICA and majority vote select the same answer on 96.1% of problems for Qwen2.5-14B (n=804; Table 10) and over 90% across all tested conditions. Varying constraint weight α across $[0, 1]$—from pure Z3 satisfiability (α=0, no source-attribution) to full source-attribution (α=1)—preserves $\geq$99% agreement (Table 11), confirming that the equivalence is driven by degeneration rather than the specific weighting scheme.

Per-trace separation analysis. Across 81% of problems with non-unanimous votes (Mistral-7B, FOLIO), per-trace constraint density is balanced between majority and minority answer groups (6.0 vs. 5.9 constraints/trace) and MAX-SAT exclusion rates are comparable (0.185 vs. 0.172): the extraction itself is unbiased. Volume amplification is $\approx$1.0×—the constraint layer neither inflates nor dampens the CoT vote ratio. Yet source-attribution binds each constraint to its originating trace's answer: 92.2% of non-unanimous problems contain cross-group shared formulas that contribute to each answer group in proportion to its trace count, providing no discriminative signal. On 82.4% of problems where the minority vote is correct, SICA still follows the majority (correction rate 17.6%). SCB is thus not an independent bias introduced by extraction but a transparency property of the pipeline: source-attribution scoring is informationally transparent to the CoT trace distribution, preserving rather than correcting the majority-vote structure.

By the data processing inequality (Cover and Thomas, 2006), $I(C_{1:K}; Y) \leq I(T_{1:K}; Y)$ for any deterministic extraction f. The contribution is not this standard bound but the identification of the $M \to T \to A \to C$ Markov chain structure in self-extraction pipelines: under the degeneration

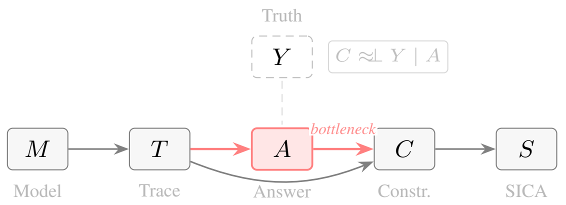

Figure 3: Causal DAG of the self-extraction pipeline. All information about ground truth Y reaching constraints C is mediated by answer A. Approximate conditional independence ($C \approx\!\perp Y \mid A$) is empirically supported by >90% argmax agreement (Observation 1) but is not a formal consequence of the DPI bound.

condition of Observation 1 ($|\mathcal{C}^{\setminus}| \approx 0$), constraints are fully determined by answers, yielding the tighter bound $I(C;Y) \leq I(A;Y)$ and the approximate conditional independence $C \approx\!\perp Y \mid A$ (Figure 3). This is a necessary condition, not a sufficient explanation; the empirical findings (Observation 1, BR, degeneration condition) provide the sufficient evidence.

Remark. Observation 1 assumes degeneration and source attribution. Under *semantic* evaluation, the equivalence additionally requires *answer alignment*: $P(c_i \text{ supports } a_i) \geq P(c_i \text{ supports } a')$ for $a' \neq a_i$, which is structurally motivated and empirically validated (BR $\in [1.66, 4.97]$; Table 2), but could fail if f systematically extracted premises contradicting their source trace's conclusion. Since Observation 1 shows SICA reduces to majority vote under degeneration, BR ≥ 1 follows from the trace-count geometry; the empirical BR consistently exceeds this floor (1.66×–4.97×; Table 2), reflecting both base-rate effects and extraction alignment.

Since the bias holds across all tested models, tasks, and extraction functions, training-free remediation strategies face a structural limitation: methods that preserve extraction accuracy necessarily inherit the model's distributional prior. This regularity holds across all 27 tested zero-shot template extraction strategies; whether few-shot or iterative refinement approaches alter this pattern remains open.

5.2 Why Remediation Fails

All 27 training-free variants within the constraint-extraction paradigm, spanning four categories, produce neutral-to-negative results (Figure 4; Appendices K–M). The pattern reveals a persistent independence-accuracy trade-off. *Diversity-seeking strategies* (temperature variation, persona sampling) shift K_{eff} by < 0.5—insufficient to

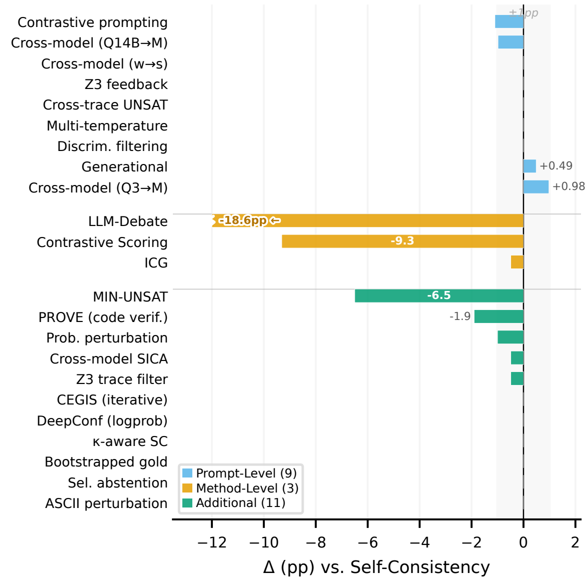

Figure 4: Twenty-three of 27 training-free strategies across four categories (four additional strategies tested but omitted for clarity). All produce neutral-to-negative Δ vs. SC. Grey band: ± 1 pp noise zone.

break correlation. *Constraint-level interventions* (explicit negation, contrastive prompting) leave the Z3 rejection rate at $\sim$3.4%—degeneration persists. *Debate* ($\kappa = 0.15$, $\Delta = -18.6$ pp) breaks correlation but collapses accuracy; *cross-model extraction* leaves κ near the same-model ceiling ($\kappa \approx 0.40$–0.47). No individual strategy achieves significance after correction (Appendix X.1). Across three model families, self-extraction cannot produce constraints independent of the distributional prior.

5.3 Diagnostic Framework

Three zero-cost diagnostics predict the outcome of any self-extraction-based aggregation method from a $K = 12$ pilot run: (1) the degenerate-regime indicator $C_0 < 5\%$ (SICA reduces to SC); (2) the diversity floor $K_{\text{eff}} \leq 1.30$ (insufficient vote diversity for any aggregation gain); (3) the bias ratio BR combined with SC regime (Type A: SC $\geq 60\%$ with BR > 1 predicts confirmation-bias degradation; Type C: SC $\leq 40\%$ predicts error amplification). These boundaries correctly characterize the failure mode across all 60 tested conditions (16 main configurations, 27 remediation variants on Mistral-7B×FOLIO, and 17 analytical conditions) within the self-extraction paradigm. The framework's practical value lies in failure-mode diagnosis—degenerate, diversity-limited, or confirmation-biased—from a K=12 pilot run, enabling practitioners to rule out self-extraction before full deployment. Conversely, when diagnostics detect sufficient capability gap

(§6), the framework correctly predicts positive gain (+6.67 pp for Mistral-7B on ProofWriter).

Decision boundaries ($C_0 < 5\%$, $K_{\text{eff}} \leq 1.30$, SC $< 40\%$) were derived from Mistral-7B and Qwen2.5-14B. We evaluate prospectively on two held-out generators: LLaMA-3.1-8B (different family) on FOLIO, where the framework predicts Type A bias and the observed $\Delta = -4.41$ pp confirms; and Qwen3-14B (same family, different generation) on FOLIO and ProofWriter, likewise confirmed ($\Delta \leq +1.47$ pp, n.s.).

6 Escaping the Bottleneck

Motivation. No training-free self-extraction method achieves independence from the model's distributional prior (§5.2). The DPI bound implies that any escape must break the self-extraction dependency: $I(C;Y) \leq I(T;Y)$ binds whenever constraints are deterministic functions of the generating model's traces. We test whether a structurally independent verifier, sharing neither architecture nor training data with the generator, provides corrective signal that escapes this bound.

Method. We use DeBERTa-large (He et al., 2021) (406M, encoder-only), fine-tuned on MultiNLI (Williams et al., 2018), to classify (premise, conclusion) pairs as entailment, contradiction, or neutral. In combo mode, each NLI-supporting vote counts w times alongside SC's K votes. We evaluate on Mistral-7B FOLIO ($n = 204$) and ProofWriter depth-5 ($n = 600$).

Zero-shot results. DeBERTa NLI alone achieves 49.51% on FOLIO and 55.00% on ProofWriter. Combo w=3 yields +6.67 pp on ProofWriter for Mistral-7B ($p < 0.001$, significant after BH correction) and +2.94 pp on FOLIO (n.s.); all generators with SC $\geq 72\%$ show no significant improvement ($\Delta \leq +1.47$ pp). Same-model SICA *degrades* ProofWriter (−2.83 pp; Table 1), illustrating that corrective signal must originate outside the model's distribution.

Capability gap. Across 108 verifier-generator-weight combinations spanning three NLI architectures, four generators, and three domains, gain decreases monotonically with generator accuracy—a *capability-gap pattern*. Of 72 above-verifier conditions, none shows a significant improvement; all significant gains concentrate below the verifier's standalone accuracy. The pattern is quantified by a perfect rank correlation across architectures ($\rho = 1.0$, $p = 0.003$, n=6; Figure 8). On ProofWriter (n=600; Table 3), only Mistral-7B (SC below DeBERTa's 55.00%) gains significantly (+6.67 pp, $p < 0.001$). At w=5, BART and RoBERTa *degrade* Qwen2.5-14B (−3.43/−4.41 pp; Appendix V).

Table 3: Cross-generator NLI escape with DeBERTa-large-mnli verifier (55.00% standalone, ProofWriter n=600). Combo gain decreases monotonically with generator capability. Only Mistral-7B (SC below the verifier) shows a significant w=3 gain.

Generator	Family	SC	Gap	Combo Δ (w=3)	p
Mistral-7B	Mistral	39.33%	+15.67	+6.67 pp	<0.001
LLaMA-3.1-8B	LLaMA	72.50%	−17.50	+0.83 pp	0.665
Qwen2.5-14B	Qwen	74.00%	−19.00	+0.67 pp	0.503
Qwen3-14B	Qwen	85.83%	−30.83	+0.50 pp	0.453

Domain-adapted verifier. Fine-tuning DeBERTa on ProofWriter depth-5 OWA (99.67% standalone) amplifies escape: +12.67/+11.67 pp for Mistral-7B/LLaMA-8B (up to 14× over zero-shot (LLaMA-8B); Table 37), extending gains to generators with SC up to ∼72%. A cost-aware cascade recovers 83–90% of gain at ∼35% cost (Appendix U). These results serve as directional evidence (the domain-adapted verifier introduces training).

Cross-model voting vs. cross-architecture NLI. An ensemble diversity control replaces the NLI verifier with a second LLM as cross voter (w=3). With Mistral-7B as cross voter on ProofWriter, all three strong generators degrade significantly (−2.33 to −9.33 pp, all $p \leq 0.020$); even a stronger cross voter (Qwen2.5-14B) degrades Qwen3-14B at w=5 (−2.50 pp, $p = 0.006$). In contrast, DeBERTa NLI at comparable standalone accuracy (55.00%) never degrades any generator ($\Delta \leq +0.83$ pp). This asymmetry reflects the structural independence of encoder-only NLI—different architecture, training objective, and input representation—from autoregressive biases shared across LLM families (Table 38).

7 Discussion and Conclusion

Our central finding is that the value of formal verification depends on the *independence* of its inputs, not the soundness of its logic. *SICA-specific*: under source-attribution scoring, SICA reduces to majority vote (Observation 1), and 27 training-free variants fail to escape. *General*: the DPI bound applies to any deterministic self-extraction, and symbolic confirmation bias—a persistent empirical pattern across all tested zero-shot self-extraction conditions, not a cognitive bias analogy—holds across nine models (7B to frontier), suggesting the bottleneck is intrinsic to self-referential constraint generation. Three diagnostics (K_{eff}, BR, C_0) correctly characterize failure modes across all tested conditions, with boundaries generalizing to held-out families (§5.3).

These results connect to ensemble diversity theory (Dietterich, 2000; Krogh and Vedelsby, 1994): with $K_{\text{eff}} \leq 3.62$ for K=12, same-model traces scale redundancy, not information. For test-time compute scaling (Snell et al., 2025), increasing K within a single model yields diminishing returns when traces share systematic biases.

Escaping the bottleneck requires structural independence (§6); cross-architecture NLI provides a proof of concept, but whether trained extractors or hybrid approaches can shift the boundary remains open. Our diagnostic framework offers a concrete tool: a K=12 pilot run with three zero-cost metrics predicts whether any self-extraction method will help before full deployment.

Limitations

Model scale. Our locally-run models range from 7B to 32B; frontier coverage (gpt-4o, gpt-4.1, gpt-5.5, o3, Gemini-2.5-Pro) relies on cross-model extraction rather than self-extraction, so we cannot confirm that self-extraction produces identical SCB patterns at frontier scale.

Domain scope. All benchmarks are English logical reasoning. FOLIO and ProofWriter represent a best-case scenario for symbolic verification: problems are designed to have clean first-order logic representations, so the observed bias ratios (1.66–4.97×) likely represent a *lower bound* on SCB in less structured domains. We excluded GSM8K and AIME because $C_0 < 5\%$ made SICA degenerate by design; whether symbolic confirmation bias manifests differently in mathematical reasoning with higher extraction rates remains untested.

Statistical power and escape scope. Statistical power is limited for effects below ~2 pp (MDE 1.86–4.84 pp across conditions); small positive effects cannot be ruled out. The capability-gap analysis rests on n=6 rank-correlation conditions (Appendix Y). Multi-seed replication uses 3–4 seeds per model; cross-seed correlation is not modeled, and a larger seed budget could tighten the variance estimates. We do not evaluate trained extractors, which could potentially break the self-extraction bound by introducing external signal.

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A Prompt Templates

SICA uses domain-specific prompt templates for constraint extraction: one for mathematical reasoning (GSM8K, AIME) and one for logical reasoning (ProofWriter, FOLIO). Both share the same system prompt. The LLM is called with temperature T=0.3, max tokens 1024, and structured JSON output mode. Up to 6 traces are processed in parallel via thread pooling, with 2 retries on failure (exponential backoff, base 0.5s).

System Prompt.

```
You are a constraint extraction tool.
    Output ONLY
valid JSON. Do NOT solve problems,
    reason about
correctness, or think step-by-step.
    Extract
constraints from the given trace
    exactly as stated.
```

Mathematical Constraint Extraction (EXTRACTION_PROMPT). Used for GSM8K and AIME datasets. The placeholder `{trace}` is replaced with the reasoning trace at runtime.

```
Extract mathematical constraints from
     this reasoning
trace as a JSON object.

Rules:
- type: "equation", "inequality", or
    "predicate"
- expression: human-readable, e.g. "x
     + y == 14"
- z3_formula: a single comparison
    expression in
  Python Z3 syntax
- source_step: step number where the
    constraint
  appears
- Use variable names from the trace

z3_formula MUST follow these rules:
- Allowed operators: ==, !=, <, >,
    <=, >=, +, -, *,
  /, **, %
- Use individual named variables: a1,
     a2, a3...
  NOT arrays a[1] or sets A
- Expand all sums explicitly:
  "a1 + a2 + a3 == 55" NOT "sum([a1,
     a2,a3])"
- NO Python builtins: sum(), len(),
    max(), min(),
  list(), set(), range(), int()
- Each z3_formula must be exactly ONE
     comparison
  (==, !=, <, >, <=, >=)
- For multiple values, write separate
     constraints
  for each

Output ONLY the JSON below, nothing
    else:
{"constraints": [{"type": "equation",
  "expression": "x + y == 14",
  "z3_formula": "x + y == 14",
  "source_step": 1}],
 "answer": "final_answer",
 "variables": ["x", "y"]}
```

```
Trace:
{trace}
```

Logical Constraint Extraction (LOGIC_EXTRACTION_PROMPT). Used for ProofWriter and FOLIO datasets.

```
IMPORTANT: You are an EXTRACTION tool
    , NOT a problem
solver.
- Do NOT re-solve or re-reason about
    the problem.
- Do NOT verify whether the trace's
    reasoning is
  correct.
- ONLY extract the logical
    constraints that are
  explicitly stated or used in the
      trace below.
- Output ONLY the JSON object,
    nothing else.

Extract logical constraints from this
     reasoning trace
about a logic problem.

The problem has facts (base
    assertions), rules
(grounded implications), and derived
    conclusions.
Convert each into a Z3 boolean
    formula.

Variable naming:
- Unary property: property_entity
  (e.g., kind_anne, red_charlie,
      round_fiona)
- Binary relation:
    relation_subject_object
  (e.g., eats_bear_squirrel,
      likes_lion_cat)
- Use lowercase with underscores only

Constraint types:
- "fact": a given assertion,
  e.g., kind_anne == True
- "rule": a grounded implication over
     specific
  entities, e.g., Implies(kind_anne,
      nice_anne)
- "derived": an inferred conclusion
    from facts +
  rules, e.g., nice_anne == True

Z3 operators:
- Implies(a, b) for if-then rules
- And(a, b, ...) for conjunctive
    conditions
- Or(a, b, ...) for disjunctive
    conditions
- Not(a) for negation
- == True / == False for boolean
    assertions

Output ONLY the JSON below, nothing
    else:
{"constraints": [{"type": "fact",
  "expression": "kind(anne)",
  "z3_formula": "kind_anne == True",
  "source_step": 1}],
 "answer": "True or False or Unknown
     ",
 "variables": ["kind_anne"]}

Trace:
{trace}
```

Output Schema. Both prompts produce a JSON object with the following structure:

```
{
  "constraints": [
    {
      "type": "<equation|inequality|
          predicate>
                or <fact|rule|derived
                   >",
      "expression": "<human-readable
          description>",
      "z3_formula": "<Z3-compatible
          expression>",
      "source_step": <integer>
    }
  ],
  "answer": "<final answer string>",
  "variables": ["<var1>", "<var2>",
      "..."]
}
```

The pipeline additionally injects an `answer_assertion` constraint per trace (not produced by the LLM):

```
{"type": "answer_assertion",
 "expression": "answer is <ans>",
 "z3_formula": "__sica_answer_<ans>
    == True",
 "source_step": -1}
```

Prompt Selection Logic. Domain assignment is determined by dataset: FOLIO and ProofWriter use LOGIC_EXTRACTION_PROMPT; GSM8K, AIME, StrategyQA, and ARC-Challenge use EXTRACTION_PROMPT. StrategyQA and ARC-Challenge are assigned to the mathematical prompt because their reasoning structure does not involve formal logic predicates or implications, despite StrategyQA producing boolean answers.

B Z3 MaxSAT Evaluation Pseudocode

Algorithm 1 describes the Z3 MaxSAT solving procedure. Algorithm 2 describes the source-trace attribution scoring.

Domain constraint variables (e.g., `x + y == 14`) and answer assertion variables (e.g., `__sica_answer_42 == True`) are logically independent in the Z3 formulation. Discriminative signal arises solely from conflicts among domain constraints: when traces supporting different answers produce mutually contradictory constraints, the solver excludes the lower-weight subset, and the scorer penalizes the corresponding answers.

Algorithm 1 Z3 Weighted Partial MAX-SAT

Require: Deduplicated constraint set $\mathcal{C} = \{(c_m, w_m, S_m)\}_{m=1}^{M}$ where c_m is a Z3 formula, w_m is weight, S_m is the set of source trace indices

Ensure: Partition $(\mathcal{C}^*, \mathcal{C}^{\setminus})$ where $\mathcal{C}^*$ is the max-weight satisfiable subset

```
opt ← Z3.OPTIMIZE()
for m = 1, ..., M do
    b_m ← fresh Boolean indicator
    opt.ADD(b_m ⇒ c_m)   ▷ soft clause guarded by indicator
    opt.ADDSOFT(b_m, weight = w_m)
end for
result ← opt.CHECK()
model ← opt.MODEL()
C* ← {(c_m, w_m, S_m) : model(b_m) = ⊤}
C\ ← C \ C*
return (C*, C\)
```

Algorithm 2 Source-Trace Attribution Scoring

Require: Satisfied set $\mathcal{C}^*$, excluded set $\mathcal{C}^{\setminus}$, candidate answers $\mathcal{A}$, trace-answer mapping τ : trace_idx $\to a$, penalty weight α

Ensure: Score $s(a)$ for each $a \in \mathcal{A}$

```
for each a ∈ A do
    T_a ← {i : τ(i) = a}   ▷ trace indices producing answer a
    pos(a) ← Σ_{(c,w,S)∈C*} w · 1[S ∩ T_a ≠ ∅]
    neg(a) ← Σ_{(c,w,S)∈C\} w · 1[S ∩ T_a ≠ ∅]
    s(a) ← pos(a) − α · neg(a)
end for
return a* = argmax_{a∈A} s(a)
```

C Capability-Accuracy Constraint Diagnostics

Conditional Correction Ratio. We define the *Conditional Correction Ratio* (CCR) to summarize verifier–generator complementarity:

$$\mathrm{CCR}(V, G) = \log \frac{P(V \text{ correct} \mid G \text{ wrong})}{P(V \text{ correct} \mid G \text{ correct})}. \tag{6}$$

CCR > 0 means V preferentially corrects G's errors; CCR < 0 indicates shared bias. CCR links our findings: (1) SCB corresponds to $\mathrm{CCR}(\mathrm{self}, G) \approx 0$; (2) Finding 2 predicts $\mathrm{CCR}(V, G) > 0$ when V is independent and capable.

Empirical CCR across verifier–generator pairs. Table 4 reports CCR for eight verifier–generator–dataset conditions, sorted by CCR descending. All three cross-architecture NLI pairs on ProofWriter yield CCR > 0 (verifier preferentially corrects generator errors), while all three same-model SICA conditions yield CCR < 0 (shared bias). CCR correlates with combo gain: Spearman $\rho = 0.839$, 95% bootstrap CI $[0.338, 0.990]$; LOO-robust (min $p = 0.014$). Four of eight conditions share the DeBERTa verifier, yielding effective $n < 10$; results should be interpreted as a consistent pattern rather than a precise effect size.

Table 4: Empirical CCR across eight verifier–generator conditions (sorted by CCR descending). CCR > 0: cross-architecture NLI verifiers on ProofWriter. CCR $<$ 0: same-model SICA or mismatched capability gap.

Generator	Verifier	Dataset	CCR	Δ (pp)
Mistral-7B	BART-MNLI	PW-600	+1.86	+5.17
Mistral-7B	DeBERTa-MNLI	PW-600	+1.67	+6.67
Mistral-7B	RoBERTa-MNLI	PW-600	+1.42	+4.17
Qwen2.5-14B	DeBERTa-MNLI	PW-600	−0.21	+0.33
Mistral-7B	DeBERTa-MNLI	FOLIO-204	−0.47	+2.94
Mistral-7B	SICA	FOLIO-204	−2.93	+3.43
Qwen2.5-14B	SICA	PW-600	−4.21	−2.50
Mistral-7B	SICA	PW-600	−4.61	−2.83

Per-condition diagnostics. Table 5 reports per-condition diagnostic metrics referenced in §M.

Model	Domain	C_0 (%)	Extr. Rate	Contr. Rate	Δ (pp)
Qwen2.5-14B	FOLIO	35.3	0.89	0.022	0.00
Qwen2.5-14B	ProofWriter	39.2	0.89	0.022	+0.33
Qwen2.5-14B	StrategyQA†	10.0	0.70	—	+3.33
Qwen2.5-14B	ARC-Chall.†	5.0	1.00	—	−1.67
Qwen2.5-7B	FOLIO+PW†	42.1	0.94	0.027	0.00
Mistral-7B	FOLIO+PW†	50.0	0.88	0.056	0.00
LLaMA-3.1-8B	StrategyQA	41.0	—	—	+2.00
LLaMA-3.1-8B	FOLIO	61.8	0.94	0.062	−4.41
DeepSeek-R1-7B	FOLIO+PW†	58.3	0.84	0.037	−3.33
QwQ-32B	Combined†	2.6	—	—	0.00
Qwen2.5-14B	GSM8K†	13.3	0.15	—	0.00

Table 5: Capability-accuracy constraint diagnostics. C_0: fraction of questions with multiple distinct candidate answers across $K = 12$ traces. Extraction rate: fraction of traces yielding at least one parseable logical constraint. Contradiction rate: fraction of extracted constraint sets with internal contradictions. Δ: SICA−SC gap (pp). †: $n \leq 60$. Dashes indicate conditions where extraction diagnostics were not computed (StrategyQA uses boolean answers without symbolic extraction; QwQ-32B Combined had near-zero C_0, making extraction analysis uninformative). This table includes additional exploratory conditions not reported in Table 1.

D Statistical Power Analysis

We compute the minimum detectable effect (MDE) for each condition at $\alpha = 0.05$ and 80% power using exact binomial power calculations over McNemar discordant pairs. Table 6 reports three grouped conditions with $n \geq 200$ (Qwen2.5-14B FOLIO and ProofWriter are pooled into a single n=804 analysis, corresponding to four Table 1 rows), which constitute our confirmatory analyses. The remaining seven conditions ($n \leq 60$) yield infinite MDE due to insufficient discordant pairs and serve only as regime characterization.

Condition	n	MDE	Obs. Δ	Obs. Pow.
Qwen-14B FOLIO+PW	804	1.86pp	+0.25pp	5.7%
LLaMA-8B StrategyQA	200	4.64pp	+2.00pp	20.6%
LLaMA-8B FOLIO	204	4.84pp	−4.41pp	18.6%

Table 6: Power analysis for confirmatory conditions ($n \geq 200$). MDE = minimum detectable effect at 80% power. All observed effects fall below their respective MDEs, and observed power remains below 21% in every case.

The largest study (Qwen-14B, n=804, 30 discordant pairs) can detect effects as small as 1.86pp, yet the observed Δ of +0.25pp is an order of magnitude smaller. The two LLaMA-8B conditions require effects of 4.64pp and 4.84pp to reach significance; neither observed Δ approaches these thresholds. All three conditions are consistent with the null hypothesis that SICA provides no accuracy gain over self-consistency.

E Bootstrap Confidence Intervals

We compute 95% confidence intervals for Δ(SICA−SC) via 10,000 bootstrap resamples (percentile method, seed 42). Table 7 reports the results for the three confirmatory conditions.

Condition	n	Obs. Δ	95% CI
Qwen-14B FOLIO+PW	804	+0.25pp	[−1.12, +1.62]
LLaMA-8B StrategyQA	200	+2.00pp	[−1.50, +5.50]
LLaMA-8B FOLIO	204	−4.41pp	[−5.88, +0.98]

Table 7: Bootstrap 95% confidence intervals for Δ(SICA − SC) in percentage points. All intervals include zero, consistent with no reliable accuracy difference.

All three intervals contain zero. The narrowest interval (Qwen-14B, [−1.12, +1.62]) rules out practically meaningful gains (>2pp) even in the most favorable case. The widest interval (LLaMA-8B FOLIO, [−5.88, +0.98]) skews negative, indicating that SICA may slightly hurt accuracy in this condition. These results corroborate the McNemar tests and power analysis: no condition provides evidence that logical constraint aggregation improves over majority voting.

F Case Studies: Discordant Predictions

Among 804 problems, SICA and SC agree on 773 (96.1%). The 31 discordant cases comprise 16 SICA wins (SICA correct, SC wrong), 14 SC wins (SC correct, SICA wrong), and 1 where both select different wrong answers. All 31 occur at exact ties (entropy=1.0, 6:6 vote split) or near-ties (entropy$\geq$0.9183, 8:4 at most), since SICA can differ from SC only when constraint weights shift the majority away from SC's tiebreak choice. We present three representative cases that illustrate the mechanism behind capability-accuracy constraint failures.

Case 1: SICA Wins (folio_134, FOLIO). *Premises*: Some mammals have teeth. Platypus have no teeth. Platypus are mammals. Humans have teeth. *Conclusion*: Humans are mammals. *Gold*: Unknown.

The premises state that *some* mammals have teeth and that humans have teeth, but "having teeth" is not sufficient for "being a mammal" under open-world semantics. SC produces a 6:6 split (Unknown:6, True:6) and tiebreaks alphabetically to True (wrong). SICA extracts 7 constraints; MaxSAT excludes 1 constraint inconsistent with the rest, reducing the weighted score for True-voting traces and tipping the aggregation to Unknown (correct). The excluded constraint likely captured an incorrect inference step (e.g., affirming the consequent) present only in True-voting traces. While SICA succeeds here, the success is coincidental: the excluded constraint happens to anti-correlate with the correct answer.

Case 2: SC Wins (ProofWriter-1066_Q6, ProofWriter). *Key chain*: White(Bob)→Round(Bob) [white things are round] → Young(Bob) [all round things are young]. *Conclusion*: "Bob is not young." *Gold*: False (Bob IS young).

SC produces a 6:6 split (False:6, True:6) and tiebreaks to False (correct). SICA extracts 17 constraints; MaxSAT excludes 2 that appear preferentially in False-voting traces. These excluded constraints likely captured correct intermediate derivations (e.g., `young_bob == True`) that appeared only in traces completing the full deduction chain. By excluding them, MaxSAT removes evidence supporting the correct answer. SICA selects True (wrong). This case illustrates how self-extracted constraints can systematically mislead: correct reasoning steps become discriminative signals that the solver mistakenly penalizes.

Case 3: Non-discriminative Constraints (ProofWriter-903_Q6, ProofWriter). *Key chain*: Nice(Mouse)→Cold(Mouse) [all nice people are cold] → Chases(Mouse,Mouse) [nice and cold → chase mouse]. *Conclusion*: "The mouse does not chase the mouse." *Gold*: False.

SC produces a 6:6 split (False:6, Unknown:6) and tiebreaks to False (correct). SICA extracts 44 constraints but excludes only 1 (2.3%). The remaining 43 are domain facts and grounded rules (`nice_mouse == True`, `Implies(nice_mouse, cold_mouse)`, etc.) that Z3 satisfies identically regardless of the final answer. Both False-voting and Unknown-voting traces share the same premises and grounded rules; they differ only in whether the trace completed the full derivation chain. The single excluded constraint carries negligible weight against 43 shared constraints, so SICA collapses to SC.

Pattern. These cases exemplify the general pattern across the 773 agreeing problems: the vast majority of extracted constraints encode shared domain structure rather than answer-specific reasoning, and MaxSAT aggregation degenerates to majority vote. Table 8 summarizes.

Table 8: Summary of three representative discordant cases. All occur at exact 6:6 vote ties. Constraint exclusion can break ties in either direction (Cases 1–2) or have no effect (Case 3).

Case	ID	Gold	SC	SICA	#Constr.	Excl. (%)
1 (SICA win)	folio_134	Unknown	True ×	Unknown ✓	7	1 (14.3%)
2 (SC win)	PW-1066_Q6	False	False ✓	True ×	17	2 (11.8%)
3 (Non-disc.)	PW-903_Q6	False	False ✓	False ✓	44	1 (2.3%)

G Oracle Control

Replacing self-extracted constraints with gold annotations (FOLIO logical formulae, ProofWriter DSL), Oracle-SICA yields +29.90 pp on Mistral-7B FOLIO ($p < 10^{-13}$), +13.73 pp on Qwen2.5-14B FOLIO ($p = 2.0 \times 10^{-7}$), and +17.17 pp on ProofWriter ($p < 10^{-14}$). Z3 agrees with gold labels on 95.1% (FOLIO) and 97.1% (ProofWriter). The MAX-SAT mechanism is sound; Mistral-7B's best self-extracted Δ (+3.43 pp) captures only 11.5% of its oracle ceiling. For Qwen2.5-14B on ProofWriter, the gap is wider still: self-extraction degrades accuracy (−2.50 pp) against an oracle ceiling of +17.17 pp, indicating that stronger models do not self-extract more effectively.

ProofWriter per-subtype analysis. ProofWriter problems encode deductive rules as Domain-Specific Language (DSL) logic programs. Each problem uses one of four predicate subtypes: attribute predicates with or without negation (AttNeg, AttNoneg), and relation predicates with or without negation (RelNeg, RelNoneg). Relation predicates involve binary relations between entities (e.g., `eats(bear, squirrel)`), while attribute predicates assign unary properties (e.g., `kind(anne)`).

The oracle pipeline parses these DSL annotations deterministically into Z3 formulas without any LLM involvement. Of the 600 ProofWriter problems, 594 (99.0%) produce valid Z3 parses; the remaining 6 fail due to unsupported DSL constructs. Across all 7,128 trace-problem pairs submitted to Z3, the solver returns a result in every case (0% solver errors). Z3 evaluations agree with ground-truth labels on 577 of 594 parseable problems (97.14%), confirming that the deterministic DSL-to-Z3 translation faithfully preserves problem semantics.

Table 9 reports the per-subtype breakdown. Relation-predicate subtypes show substantially larger gains than attribute-predicate subtypes: RelNeg improves by +28.58 pp and RelNoneg by +23.29 pp, compared to +10.00 pp for AttNeg and +8.44 pp for AttNoneg. Relation predicates involve longer reasoning chains with more intermediate entities, producing greater answer diversity across traces (C_0) and more opportunities for oracle constraints to correct errors.

Table 9: ProofWriter oracle experiment per-subtype breakdown. Oracle-SICA uses gold DSL annotations parsed deterministically to Z3 for consistency checking. Relation-predicate subtypes (RelNeg, RelNoneg) show larger gains than attribute-predicate subtypes; relation subtypes involve longer reasoning chains and benefit more from oracle constraints. Bold indicates best accuracy per row.

Subtype	n	SC Acc (%)	Oracle Acc (%)	Δ (pp)
AttNoneg	154	90.26	**98.70**	+8.44
AttNeg	160	81.88	**91.88**	+10.00
RelNoneg	146	60.96	**84.25**	+23.29
RelNeg	140	60.71	**89.29**	+28.58
Overall	600	74.00	**91.17**	+17.17

H Constraint Ablation Details

Table 10: Constraint ablation (Qwen2.5-14B, $n = 804$). Agreement: predictions identical to SC. MAX-SAT and count-only produce nearly identical predictions, confirming that self-extracted constraints carry no signal beyond vote counts.

Aggregation	Acc. (%)	Agreement w/ SC
SC (majority vote)	76.24	—
SICA (MAX-SAT)	76.49	773 / 804
Count-only	76.62	774 / 804
Random-weight	75.87	764 / 804

Table 11: MAX-SAT vs. count-only agreement across constraint weight $\alpha \in [0, 1]$ on FOLIO ($n = 204$). Mistral-7B (BR $= 4.97\times$, seed 42); Qwen2.5-14B (BR $= 3.85\times$, seed 123). α=0: pure Z3 satisfiability (no source-attribution); α=1: full source-attribution penalty. Agreement $\geq$ 99% at all α including endpoints, confirming vacuity is weight-insensitive.

α	Mistral Agr. (%)	Qwen2.5 Agr. (%)
0.0	100	100
0.1	100	100
0.3	100	100
0.5	100	100
0.7	99	100
0.9	99	100
1.0	99	100

I Remediation Strategy Details

Table 12: Nine prompt-level remediation strategies. Baseline: Qwen2.5-14B FOLIO, SC = 75.98%, $\Delta = -0.49$ pp. None reaches significance. Three method-level strategies are reported in Table 13.

Strategy	Target	Δ (pp)	Key finding
Contrastive[a]	Prompt	−1.10	non-disc. 6.0% (↓ from 8.2%)
Cross-model[b]	Architecture	0.00	disc. 0% (weak→strong); ±1 pp (strong→weak)
Qwen3-14B[c]	Capability	−0.49	$p = 0.655$
Z3 feedback	Solver	—	0.9% UNSAT trigger
Cons. filtering	Noise	0.00	$T \geq 4 \rightarrow$ SC
Cross-trace UNSAT	Independence	—	56.4% $\emptyset$ vars
T=1.0[d]	Diversity	0.00	isolated peak at T=0.7 only
Multi-temp[e]	Diversity	0.00	b=5, c=5 (balanced)
Disc. filtering[f]	Filtering	0.00	803/804 agree w/ SICA

[a] n=182 (extraction failures excluded). [b] Weak→strong: Mistral-7B extracts from Qwen2.5-14B traces. Strong→weak: Qwen3-14B (60.29%) and Qwen2.5-14B (58.33%) extract from Mistral-7B FOLIO traces (n=204); self-extraction baseline 59.31%. [c] n=204, SC = 83.33%. [d] Mistral-7B FOLIO, n=204. [e] Mistral-7B FOLIO, $T \in \{0.3, 0.7, 1.2\}$, 4 traces each, n=204. [f] Qwen2.5-14B, n=804; removing non-discriminative constraints.

J Additional Intervention Results

K Extraction Error Taxonomy

Constraints must discriminate between candidate answers for MAX-SAT to produce a ranking different from majority vote. We measure discriminative power at two levels on Qwen2.5-14B across FOLIO and ProofWriter ($n = 804$). At the source-trace level, 39.2% of total constraint weight originates from traces whose answer differs from the plurality, but only 12.1% of individual constraints fall into this minority category. At the Z3 semantic level, 91.8% of constraints [95% CI: 91.2, 92.4] are technically discriminative, producing different satisfiability outcomes when evaluated against different candidate answers. Only 8.2% are non-discriminative.

The high discriminative rate does not translate into accuracy gains because discrimination mirrors the model's own answer distribution. Each trace produces constraints with its own variable names, so Z3 treats identically-intended propositions as unrelated symbols; among cross-answer constraint pairs, 56.4% share zero Z3 variables. This variable isolation means constraints from different traces are trivially jointly satisfiable, and MAX-SAT retains nearly all of them regardless of source answer. Removing the 8.2% non-discriminative constraints yields zero accuracy improvement. The discriminative majority does not provide independent signal; it reinforces the majority answer that SC would already select. This confirmation bias, not a lack of discrimination, explains why SICA degenerates to majority voting (Appendix H).

Table 13: Method-level interventions on Mistral-7B with McNemar's paired test. All produce neutral-to-catastrophic results. ICG shows two FOLIO seeds; p for the first. *** $p < 0.001$; ** $p < 0.01$.

		FOLIO ($n = 204$)		LogiQA ($n = 200$)	
Method	Target	Δ (pp)	p	Δ (pp)	p
SC baseline	—	0.00	—	0.00	—
ICG	Trace coupling	−1.96 / +1.47	.206	−0.50	.706
Contrastive	Scoring bias	**−9.31**	.007	−4.50	.160
Debate	Role diversity	**−18.63**	<.001	**−23.00**	<.001

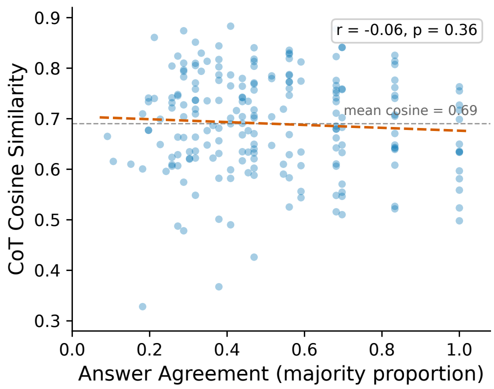

Figure 5: Answer agreement vs. CoT cosine similarity (Mistral-7B, FOLIO, n=204). Flat regression ($r = -0.06$, $p = 0.36$); stronger models show significant positive correlation ($r = +0.24$, Qwen-14B; Table 21).

Table 14: Additional intervention results beyond the nine prompt-level and three method-level strategies reported in the main text. All produce neutral-to-negative results, completing the exhaustive remediation analysis across 27 distinct approaches.

#	Method	Type	Target	Δ (pp)	Key Finding
13	PROVE (code verif.)	Paradigm	Formal proof	-1.9	15.2% compilable; conclusive acc. = 50%
14	CEGIS (iterative)	Multi-round	Consistency	$0.0^{\dagger}$	R1 $+3.43$ pp then oscillates
15	DeepConf (logprob)	Filtering	Confidence	0.0	$\Delta_{\text{logprob}} = 0.006$ (no separation)
16	Cross-model SICA	Cross-model	Independence	-0.49	κ_{cross}=0.42 vs κ_{within}=0.53
17	Problem perturbation	Input diversity	Trace diversity	<0	κ *increased* to 0.70 (anti-effect)
18	MIN-UNSAT	Solver objective	Constraint sel.	-6.5	Mean excluded constraints = 1.3
19	κ-aware SC	Voting rule	Weighted voting	0.0	11/11 conditions $\Delta = 0$
20	Z3 trace filter	Trace filtering	Quality	-0.49	Precision 49.1%, 7 questions affected
21	Bootstrapped gold	Iterative	Self-improvement	$0.0^{\dagger}$	R1 $+2.45$ pp then oscillates
22	Selective abstention	Diagnostic	Confidence	$+4.4^{*}$	AUROC = 0.673, at 80% coverage
23	ASCII perturbation	Input diversity	Format diversity	0.0	κ: 0.39 $\rightarrow$ 0.36 (negligible)
24	Atom pipeline	Paradigm	Decomposition	-10.0	Atomic extraction degrades
25	Premise-only filter	Constraint sel.	Bias isolation	$+2.45$	$p = 0.359$; halves conclusion-bias
26	DPP trace selection	Trace diversity	Determinantal	$+1.47$	$p = 0.629$; K_{sel}=8
27	Z3 strict-mode	Solver config	UNSAT filtering	0.0	McNemar b=1, c=1; 63% UNSAT

All experiments on Mistral-7B FOLIO (n=204) unless noted. † Oscillates without convergence; net effect ≈ 0. * Accuracy gain only at reduced coverage (diagnostic value, not a general improvement). Methods 1–12 are reported in Table 12 and Table 13.

L Hyperparameter Sensitivity

L.1 K Sensitivity

Figure 6 shows SICA Δ as a function of the number of sampled traces K for Mistral-7B on FOLIO ($n = 204$, no fixed seed). The relationship is non-monotonic: $K = 4$ yields $\Delta = -1.47$ pp, $K = 8$ yields $\Delta = -1.96$ pp, $K = 12$ reaches $\Delta = +3.43$ pp, $K = 16$ drops to $\Delta = +1.47$ pp, and $K = 20$ partially recovers to $\Delta = +1.96$ pp. The recovery at $K = 20$ rules out a simple peaked pattern, but $K = 12$ remains the only setting with $\Delta > 2$ pp ($p = 0.092$, marginal).

The apparent Δ peak at $K = 12$ is confounded with SC baseline. SICA absolute accuracy plateaus at 59.31% (121/204) for $K \geq 12$: identical at $K = 12$ and $K = 20$. The Δ variation across these settings is driven primarily by SC fluctuations rather than by SICA improvement. $K = 12$ produces a relatively low SC (55.88%), placing it in the intermediate-SC region; $K = 8$ (SC = 58.33%) and $K = 20$ (SC = 57.35%) have higher baselines, compressing Δ even though SICA accuracy does not change for $K \geq 12$. Additional traces beyond $K = 12$ provide no benefit to SICA absolute performance, consistent with the constraint quality bottleneck. The $K = 12$ Δ captures only 25.0% of the oracle ceiling ($+13.73$ pp).

L.2 Temperature Sensitivity

Temperature sensitivity on Mistral-7B FOLIO ($n = 204$, default seed) reveals a sharp isolated peak rather than a smooth trend. Across four temperature settings, only $T = 0.7$ produces a positive effect: $T = 0.3$ yields $\Delta = -1.47$ pp (SC = 58.33%), $T = 0.5$ yields $\Delta = -0.98$ pp (SC = 58.33%), $T = 0.7$ yields $\Delta = +3.43$ pp (SC = 55.88%), and $T = 1.0$ yields $\Delta = 0.00$ pp (SC = 55.39%). All other temperatures yield negative or null effects, and the single positive result at $T = 0.7$ is confounded with SC baseline. Combined with the seed sensitivity (0 of 4 seeds reaching $p < 0.05$ at $T = 0.7$), the positive trend depends on a specific combination of temperature, seed, and baseline accuracy that same-model sampling rarely achieves.

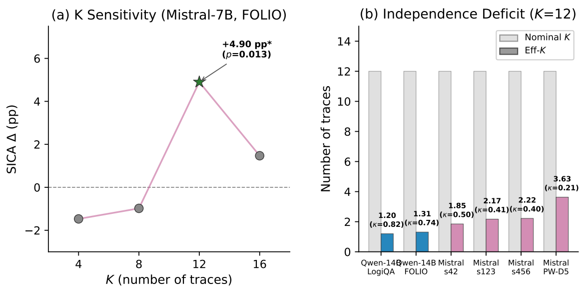

Figure 6: SICA Δ (pp) vs. number of traces K for Mistral-7B on FOLIO ($n = 204$, default seed, $T = 0.7$). Non-monotonic pattern: SICA accuracy plateaus at 59.31% for $K \geq 12$; the apparent Δ peak at $K = 12$ partly reflects a relatively low SC baseline (55.88%) at that point. Error bars: 95% bootstrap CI.

The parallel between K and T sensitivity points to a common underlying variable. In both sweeps, the setting that produces the largest SICA gain also produces a relatively low SC baseline (55.88%), which places it in the intermediate-SC region. The hyperparameters themselves (K, T) are not the causal driver; they modulate SC level, and it is the SC level that determines whether SICA has operating room.

M Entropy Decomposition

We partition problems by trace agreement: high-entropy (majority count ≤ 8 out of $K = 12$) and low-entropy (majority count ≥ 9). Table 15 reports SICA Δ within each partition across eight model-dataset conditions.

Table 15: Entropy decomposition of SICA Δ (pp). High-entropy: majority count ≤ 8; low-entropy: majority count ≥ 9. †Partial data (180/600). Pre-tiebreaking SC values; pattern robust to D064 correction.

Model	Dataset	High-entropy		Low-entropy	
		n	Δ	n	Δ
Mistral-7B	FOLIO	78	**+12.8**	126	0.0
Mistral-7B (s123)	FOLIO	77	**+9.1**	127	0.0
Mistral-7B (s456)	FOLIO	75	−1.3	129	0.0
Mistral-7B	PW-D5	267	−4.9	333	−1.2
Qwen2.5-14B	FOLIO	37	−8.2	167	−0.6
Qwen2.5-14B	LogiQA	23	−4.3	177	0.0
Qwen2.5-7B	FOLIO	47	−2.1	157	0.0
LLaMA-3.1-8B†	PW-D5	76	−3.9	104	−1.9

Two patterns emerge. First, low-entropy problems produce $\Delta \approx 0$ pp across all eight conditions: when traces agree, constraints carry no information beyond the majority vote.

Second, the direction of SICA's effect on high-entropy problems depends on the accuracy regime, not on entropy alone. In the intermediate-SC region (Mistral-7B on FOLIO, default-seed and seed-123 runs), high-entropy Δ reaches $+12.8$ and $+9.1$ pp: traces genuinely disagree, and constraints from correct-answer traces provide discriminative signal that resolves the disagreement. In Type A conditions (Qwen2.5-14B on FOLIO, $\Delta = -8.2$ pp), high-entropy problems represent rare deviations from a strong model's consensus; constraints inherit the model's confirmation bias. In Type C (Mistral-7B on ProofWriter, $\Delta = -4.9$ pp), high-entropy problems reflect systematic errors. The mechanism is not entropy per se but the interaction of entropy with model capability.

N Failure Mode Analysis

The regime-dependent pattern in Table 1 reflects three distinct failure modes of self-extracted logical constraints.

Type A: confirmation bias (SC $\geq \sim$60%). Strong models produce consistent reasoning traces, and constraints extracted from those traces mirror the answer distribution. When most constraints support the same candidate, MAX-SAT excludes few clauses ($|\mathcal{C}^{\setminus}| \approx 0$), the penalty term in Eq. 3 vanishes, and the scoring function degenerates to weighted vote counting (Eq. 4). SICA and SC then produce identical predictions. The ablation in Appendix H confirms this empirically: on Qwen2.5-14B (FOLIO + ProofWriter, $n = 804$), MAX-SAT and count-only baselines agree on 803 of 804 predictions.

Type B: non-formalizable domains. LogiQA problems require pragmatic and contextual reasoning that resists clean logical formalization. Qwen2.5-14B on LogiQA achieves an extraction rate of 86.4%, but the resulting constraints have a contradiction rate of only 0.8%: extracted formulas rarely conflict across traces, leaving MAX-SAT with no logical tension to exploit. Mistral-7B on LogiQA sits in the same SC range (50.00%) yet produces $\Delta = +1.50$ pp ($p = 0.581$); Qwen2.5-14B on the same domain yields $\Delta = -1.00$ pp ($p = 0.625$). Neither model benefits, confirming that the right accuracy regime is necessary but not sufficient when the domain resists formalization.

Type C: error amplification (SC $\leq \sim$40%). When model accuracy is low, the majority of traces reach incorrect answers. Constraints extracted from wrong traces encode incorrect reasoning as formal rules. Z3 treats these rules as soft clauses with weight proportional to their frequency, so the solver preferentially satisfies the wrong constraints and penalizes traces that happen to reach the correct answer. Mistral-7B on ProofWriter ($\Delta = -2.83$ pp, raw $p = 0.014$; BH adj. $p = 0.042$) demonstrates this effect: SICA converts a weak positive signal in trace diversity into a net negative by formalizing the errors.

O Cross-Model Confirmation Bias

Table 16 reports MaxSAT-weighted confirmation bias metrics across three model families on FOLIO ($n = 204$). Weighted bias ratio and wrong-supporting constraint percentage both decrease with model capability, while SICA's accuracy effect shifts from positive (Mistral, intermediate-SC regime) to negative (LLaMA, high-SC regime). Raw bias ratios (4.97×, 3.85×, 2.88×) are reported in §5.1. LLaMA weighted BR CI: [2.19, 3.28].

Frontier model universality. Table 2 (main text, §5.1) extends confirmation bias analysis to nine

Table 16: Cross-model confirmation bias on FOLIO ($n = 204$, $K = 12$). BR (wtd.): MaxSAT-weighted ratio of error-supporting constraints on incorrect vs. correct problems. Wrong %: weighted fraction of constraints supporting the incorrect majority on error problems. Δ: SICA − SC (pp).

Model	BR (wtd.)	Wrong (%)	Δ (pp)
Mistral-7B	4.24×	80.9	+4.90
Qwen2.5-14B	3.69×	78.7	−0.98
LLaMA-3.1-8B	2.66×	72.7	−4.41

models spanning 7B open-source to frontier closed-source. All models exhibit BR > 1 (range: 1.66–4.97×). Extraction context varies: open-source models use self-extraction (the model extracts constraints from its own traces); frontier models use cross-model extraction (Mistral-7B extracts FOL from the target model's traces), as closed-source APIs do not expose internal representations suitable for direct symbolic extraction.

ProcessSim definition. Process similarity is the mean pairwise cosine similarity of TF-IDF trace representations:

$$\text{ProcessSim}(q) = \frac{2}{K(K-1)} \sum_{i<j} \cos(\mathbf{v}_i^q, \mathbf{v}_j^q), \tag{7}$$

where $\mathbf{v}_i^q$ is the TF-IDF vector of trace t_i for question q.

Process-κ gradient. Under cross-model extraction (Mistral-7B extracts FOL from other models' traces), SICA gain decreases monotonically with process-level cosine similarity (Table 17; Spearman $\rho = -1.0$, $n = 5$, one-tailed $p = 0.0083$). Gain requires two conditions: diverse traces (low κ_{proc}) *and* substantial self-bias (high BR). R1-Distill-8B satisfies both ($\kappa_{\text{proc}} = 0.248$, BR $= 2.92$) and gains +2.45 pp; o3 has moderate trace diversity ($\kappa_{\text{proc}} = 0.596$) but near-minimal bias (BR $= 1.66$, exhibiting a bimodal error pattern: 82% of error problems have BR $= \infty$ but low finite-BR), yielding $\Delta = 0$.

Table 17: Process-κ gradient under cross-model extraction (Mistral-7B → target model traces, FOLIO). κ_{proc}: process-level cosine similarity. Δ: SICA − SC (pp). Spearman $\rho(\kappa_{\text{proc}} \to \Delta) = -1.0$, $p = 0.0083$. *Interim ($n = 95/204$).

Model	κ_{proc}	Δ (pp)	BR
R1-Distill-8B	0.248	+2.45	2.92
gpt-5.5	0.563	+0.49	3.48
o3	0.596	0.00	1.66
Gemini-2.5-Pro*	0.790	−1.05	2.79
gpt-4.1	0.792	−1.47	2.70

P Independence-Accuracy Scatter Plot

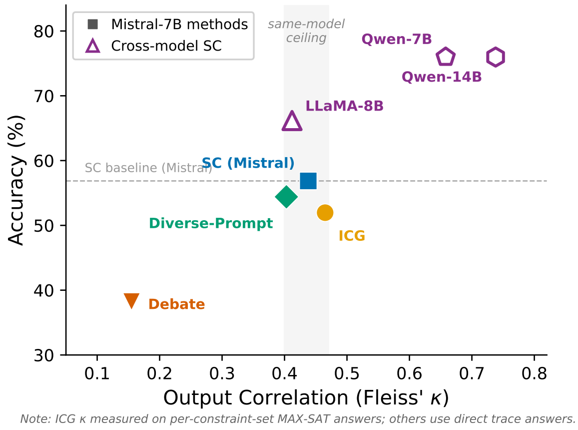

Figure 7: Independence-accuracy trade-off across aggregation methods (Mistral-7B, filled markers) and model families (LLaMA-3.1-8B, Qwen2.5-7B, Qwen2.5-14B; open markers) on FOLIO ($n = 204$). Same-model methods cluster at $\kappa \approx 0.40$–0.47 (gray band), while cross-model SC shows stronger correlations ($\kappa = 0.41$–0.74) at higher accuracy. Debate breaks correlation ($\kappa = 0.15$) at catastrophic accuracy cost. Dashed line: Mistral-7B SC baseline.

Table 18: Independence-accuracy trade-off (FOLIO, $n = 204$). Top: aggregation methods on Mistral-7B; bottom: SC across three model families. All families produce $\kappa \geq 0.41$, confirming an architecture-independent ceiling. κ: Fleiss' kappa on raw answers.

Model	Aggregation	κ	Eff-K	K	Acc. (%)
Mistral-7B	Debate	0.15	4.44	12	38.24
	Diverse-Prompt	0.40	2.21	12	54.41
	Self-Consistency	0.44	2.02	10	56.86
	ICG	0.47	1.93	10	51.96
LLaMA-3.1-8B	Self-Consistency	0.41	2.17	12	66.18
Qwen2.5-7B	Self-Consistency	0.66	1.46	12	75.98
Qwen2.5-14B	Self-Consistency	0.74	1.32	12	75.98

Trace independence metrics. Table 19 reports the full 15-condition breakdown of Fleiss' κ and Effective-K at $K = 12$, referenced in §3.1.

Q Cross-Model Kappa Analysis

Table 20 compares trace independence across three model families on the same benchmarks. On FOLIO ($n = 204$), all four models produce $\kappa \geq 0.41$: LLaMA-3.1-8B ($\kappa = 0.41$), Mistral-7B ($\kappa = 0.44$), Qwen2.5-7B ($\kappa = 0.66$), and Qwen2.5-14B ($\kappa = 0.74$). The bottleneck is architecture-independent—no family escapes the correlation floor. Within the Qwen family, κ increases from 0.66 (7B) to 0.74 (14B), but Mistral-7B (7B) and LLaMA-3.1-8B (8B) differ by $\Delta\kappa = 0.03$ despite belonging to different families, suggesting that

Table 19: Trace independence metrics ($K = 12$, sorted by κ descending). κ: Fleiss' Kappa. K_{eff}: Effective-K (Eq. 1). SC: Self-Consistency baseline. Δ: SICA − SC (pp). $*p < 0.05$ (uncorrected; BH adj. $p = 0.042$, m=6); positive Δ values do not reach significance. multi-T = $T \in \{0.3, 0.7, 1.2\}$, 4 traces each.

Model	Condition	κ	K_{eff}	SC (%)	Δ (pp)
Qwen3 (think)	FOLIO	0.91	1.09	86.27	+0.49
Qwen2.5-14B	LogiQA	0.82	1.20	79.00	−1.00
Qwen3	FOLIO	0.77	1.26	83.33	+1.47
Qwen2.5-14B	FOLIO	0.74	1.30	75.98	−0.49
Qwen2.5-7B	FOLIO	0.66	1.46	75.98	+0.49
Mistral-7B (T=0.3)	FOLIO	0.53	1.77	58.33	−1.47
Mistral-7B	LogiQA	0.51	1.81	50.00	+1.50
Mistral-7B (s456)	FOLIO	0.51	1.82	55.88	0.00
Mistral-7B	FOLIO	0.50	1.84	55.88	+3.43
Mistral-7B (s123)	FOLIO	0.49	1.88	54.41	+2.94
Mistral-7B (s789)	FOLIO	0.45	2.03	59.31	−2.45
Mistral-7B (T=0.5)	FOLIO	0.52	1.80	58.33	−0.98
Mistral-7B (multi-T)	FOLIO	0.48	1.91	57.84	0.00
Mistral-7B (T=1.0)	FOLIO	0.48	1.91	55.39	0.00
Mistral-7B	PW-D5	0.21	3.62	39.33	**−2.83***

cross-family variation ($\Delta\kappa$ up to 0.22 at similar scale) exceeds within-family scale effects ($\Delta\kappa = 0.08$). Disentangling scale from architecture requires controlled same-family experiments across $\geq$3 scales.

Table 20: Cross-model trace independence ($K = 12$, SC aggregation, FOLIO + ProofWriter). Higher κ indicates stronger correlation and lower effective sample size.

Model	Dataset	κ	K_{eff}	SC (%)
Qwen2.5-14B	FOLIO	0.74	1.32	75.98
Qwen2.5-7B	FOLIO	0.66	1.46	75.98
Mistral-7B	FOLIO	0.44	2.02	56.86
LLaMA-3.1-8B	FOLIO	0.41	2.17	66.18
Qwen2.5-14B	PW-D5	0.62	1.54	—
Mistral-7B	PW-D5	0.21	3.62	39.33

R Cross-Model Process-Level Independence

Table 21 extends the process-level independence analysis (§5.1) to four model families on FOLIO ($n = 204$). Mean pairwise cosine similarity of TF-IDF trace representations exceeds answer-level κ in every family, but the gap is model-strength dependent: it narrows from 0.31 (LLaMA) and 0.23 (Mistral) to 0.02 (Qwen-14B). For weaker models ($r \approx 0$), answer diversity does not reflect reasoning diversity—process-convergent but answer-divergent trace pairs constitute 9.5% of all pairs on Mistral-7B versus 2.5–3.2% on other families. For stronger models ($r \approx 0.23$, $p \leq 0.001$), process and answer agreement align, and answer-level κ already captures the bottleneck. Process-κ thus provides a crucial additional diagnostic specifically for weaker models, where answer-level metrics underestimate reasoning homogeneity.

Table 21: Cross-model process-level independence on FOLIO ($n = 204$, $K = 12$). Mean Cosine: mean pairwise TF-IDF cosine similarity of reasoning traces (Eq. 7). Answer κ: per-question answer agreement. Gap: Cosine − κ. r: Pearson correlation between ProcessSim and answer agreement. Conv.: fraction of process-convergent but answer-divergent trace pairs. $^{**}p < 0.01$; $^{***}p < 0.001$.

Model	Mean Cos.	Ans. κ	Gap	r	p	Conv. (%)
Mistral-7B	0.690	0.460	0.230	−0.064	0.362	9.5
LLaMA-3.1-8B	0.759	0.447	0.312	−0.001	0.985	3.2
Qwen2.5-7B	0.790	0.658	0.132	+0.226	0.001**	2.8
Qwen2.5-14B	0.766	0.744	0.022	+0.244	<0.001***	2.5

Representation robustness. Replacing TF-IDF with SBERT embeddings (all-MiniLM-L6-v2) yields ProcessSim = 0.91 (vs. TF-IDF 0.74), widening the gap over answer-level $\kappa = 0.55$. The process homogeneity finding is robust to representation choice.

S Constraint Quality Across Methods

Table 22 compares constraint quality metrics across the three method-level interventions on Mistral-7B FOLIO ($n = 204$). Standard SICA extracts from reasoning traces and achieves the highest parse rate (97.1%) but also the highest redundancy (42.9%): nearly half of all constraints are syntactic duplicates across the 12 traces, quantifying the co-sourcing effect of same-model generation. ICG generates constraints directly from the problem statement without trace conditioning. Its lower parse rate (74.6%) reflects the difficulty of standalone formalization at the 7B scale, though successfully parsed constraints compile at a higher rate (99.4% vs. 96.5%). ICG produces fewer unique constraints per question (24.9 vs. 39.3) with lower redundancy (32.3%), consistent with the absence of trace-level repetition. Contrastive scoring generates pro and con constraint sets for each candidate answer. The format yields far fewer unique constraints per question (9.2) with zero redundancy (each pro/con pair is structurally distinct), but 62.7% of constraints are trivially satisfied by Z3—they impose no logical tension and cannot discriminate between candidates. Only 51.0% of Contrastive constraints are discriminative, compared to 81.4% for standard SICA and 77.0% for ICG.

Per-trace constraint yield uniformity. Observation 1 assumes approximately uniform constraint

Table 22: Constraint quality metrics across method-level interventions (Mistral-7B, FOLIO, $n = 204$). Parse: fraction of extraction attempts yielding valid JSON. Z3 Compile: fraction compiling to Z3 formulas. Unique/q: mean unique constraints per question. Redundancy: fraction of constraints duplicated across traces. Disc.: fraction discriminative (different Z3 outcome per candidate). Triv. Sat.: fraction trivially satisfied (no logical tension).

Method	Parse	Z3	Uniq/q	Redund.	Disc.	Triv. Sat.
Standard SICA	97.1%	96.5%	39.3	42.9%	81.4%	—
ICG	74.6%	99.4%	24.9	32.3%	77.0%	—
Contrastive	74.0%	94.4%	9.2	0.0%	51.0%	62.7%

yield per trace within each answer group. We verify this on Mistral-7B FOLIO (K=12, n=204). Per-problem yield CV (coefficient of variation across traces within each problem) has median 0.257 and mean 0.319; 64.2% of problems fall below CV $= 0.3$, and 91.2% below 0.5, supporting the uniformity assumption. Majority-answer traces average 5.61 constraints and minority-answer traces 6.04 (Mann–Whitney $p = 5.2 \times 10^{-6}$); correct-answer traces average 5.41 and incorrect 6.05 ($p = 5.3 \times 10^{-11}$). Although statistically significant, the absolute difference ($\sim$0.6 constraints, $< 12\%$) is insufficient to alter the argmax-preservation result: SICA and SC select the same answer on $>$96% of problems regardless of yield variation.

T Error Categorization of Method-Level Interventions

We categorize discordant predictions (where the method and SC disagree) for Debate and Contrastive on FOLIO ($n = 204$) to understand why these interventions degrade accuracy.

Debate discordant analysis. Debate produces 60 correct→wrong flips and 22 wrong→correct flips on FOLIO. Answer-type breakdown reveals that False-labeled questions are most vulnerable: 82.8% of False questions that SC answers correctly are flipped by Debate, compared to 63.9% for True and 25.5% for Unknown. Counterintuitively, failures concentrate not where debate breaks down but where it succeeds most thoroughly: discordant questions average 2.93 effective debate rounds (proposer-critic exchanges producing substantive updates) versus 2.52 for concordant ones. More extensive same-model debate produces more confident but incorrect constraints, consistent with the independence limitation—the critic draws on the same model distribution as the proposer, reinforcing shared biases rather than correcting them.

Contrastive discordant analysis. Contrastive produces 34 discordant questions on FOLIO. All 34 produced non-degenerate scores (both pro and con constraint sets non-empty), compared to 20.1% degenerate scores in the full sample. This indicates that scoring failures on discordant questions stem from systematic pro-bias rather than missing constraints. Among discordant questions, 97.1% contain negation words in the problem statement versus 76.5% in the full sample, suggesting that negation-bearing problems are particularly susceptible to pro-constraint imbalance. The average pro/con score ratio is 1.0 for discordant questions versus 0.68 for the full sample: the method generates balanced-looking scores that mask directional errors.

Cross-method overlap. Of the 79 unique questions on which either method disagrees with SC, only 15 overlap between Debate and Contrastive (Jaccard index $= 0.19$). All 15 shared discordant questions contain negation words. The low overlap confirms that the two methods attack different vulnerability types—Debate through extended same-model argumentation that amplifies confident errors, Contrastive through pro-bias in constraint generation—while sharing a common root cause in correlated same-model voting: both methods rely on the same model's output distribution and cannot introduce genuinely independent signal.

U TOST Equivalence Tests

We apply two one-sided tests (TOST) (Lakens, 2017) with equivalence margin ± 2 pp to all 16 non-degenerate SICA–SC comparisons, classifying each condition as *equivalent* (TOST significant), *genuinely different* (McNemar $p < 0.05$ and TOST not significant), or *underpowered* (neither test significant). Four conditions are equivalent, confirming that SICA and SC produce statistically indistinguishable accuracy. Two conditions show genuine differences, both negative: Mistral-7B ProofWriter (-2.83 pp, $p = 0.014$) and Qwen2.5-14B ProofWriter (-2.50 pp, $p = 0.011$). The remaining ten conditions are underpowered, reflecting the limited statistical power at n=200–204 for detecting $|\Delta| \leq 2$ pp effects.

Margin sensitivity. Table 23 shows that the classification is robust across margins: regardless of the chosen margin, no condition shows a significant positive SICA effect.

Table 23: TOST classification sensitivity across equivalence margins (16 non-degenerate conditions).

Margin	Equivalent	Different	Underpowered
± 1 pp	0	2	14
± 2 pp	4	2	10
± 3 pp	10	0	6

Relative-margin TOST. The absolute ± 2 pp margin is more stringent at high baselines ($\pm 2.4\%$ relative at SC $= 85\%$) than low ones ($\pm 5.1\%$ at SC $= 39\%$). A relative $\pm 5\%$ of SC margin addresses this asymmetry: 8 conditions are equivalent, 2 genuinely different (both negative), and 6 underpowered. The additional four equivalent conditions are all high-SC ($\geq 75\%$), where the wider relative band captures practical indistinguishability. No condition shows a significant positive effect under either margin.

Table 24: Conditional Correction Ratio across eight verifier–generator conditions. CCR > 0: cross-arch NLI on ProofWriter. CCR < 0: same-model SICA or weak capability gap.

Generator–Verifier	Dataset	CCR	Δ (pp)
Mistral $\times$ BART-NLI	PW	$+1.86$	$+5.17$
Mistral $\times$ DeBERTa-NLI	PW	$+1.67$	$+6.67$
Mistral $\times$ RoBERTa-NLI	PW	$+1.42$	$+4.17$
Qwen-14B $\times$ DeBERTa-NLI	PW	-0.21	$+0.33$
Mistral $\times$ DeBERTa-NLI	FOLIO	-0.47	$+2.94$
Mistral $\times$ SICA	FOLIO	-2.93	$+3.43$
Qwen-14B $\times$ SICA	PW	-4.21	-2.50
Mistral $\times$ SICA	PW	-4.61	-2.83

Conditional Correction Ratio data.

Disagreement-Routed Cascade details. DRC invokes NLI only when SC consensus falls below threshold τ, avoiding unnecessary verifier calls on high-agreement problems. Gains concentrate on weak generators (SC $< 40\%$), consistent with the capability-gap hypothesis (§6). Regime-dependent bottleneck analysis appears in Appendix W.

Table 25: DRC on Mistral-7B (K=12, DeBERTa-large-mnli, w=3). τ=0.7 recovers 83–90% of full-combo gain at $\sim$35% NLI cost across both datasets.

Dataset	Method	Acc.	NLI Calls	Cost	Recovery
PW-D5	SC only	39.33%	0/600	0%	—
	DRC τ=0.7	**45.33%**	**207/600**	**34.5%**	**90.0%**
	Full combo	46.00%	600/600	100%	100%
FOLIO	SC only	55.88%	0/204	0%	—
	DRC τ=0.7	**58.33%**	**71/204**	**34.8%**	**83.3%**
	Full combo	58.82%	204/204	100%	100%

Table 26: Cross-architecture NLI combo on ProofWriter depth-5 (n=600, Mistral-7B traces, SC $=$ 39.33%). Three NLI verifiers $\times$ three weight settings. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (McNemar).

Verifier	Standalone	w	Δ (pp)	p
DeBERTa-MNLI	55.00%	1	$+0.83$	n.s.
		3	$+6.67$	$<$0.001***
		5	$+11.00$	$<$0.001***
RoBERTa-MNLI	37.67%	1	-0.17	n.s.
		3	$+4.17$	0.003**
		5	$+4.33$	0.037*
BART-MNLI	46.17%	1	0.00	n.s.
		3	$+5.17$	$<$0.001***
		5	$+6.33$	0.003**

All verifiers fine-tuned on MultiNLI. Unknown recall $<$3.5% across all verifiers.

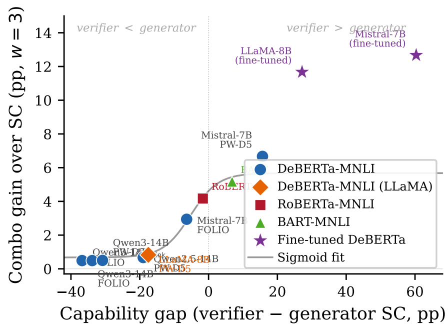

Figure 8: Combo gain vs. capability gap (verifier standalone $-$ generator SC) at w=3 across three NLI architectures. Gain decreases monotonically (Spearman $\rho = 1.0$, exact permutation $p = 0.003$, n=6; four of six conditions share the DeBERTa verifier—see Limitations); significant gains occur only for generators below the verifier's standalone accuracy. Stars: domain-adapted DeBERTa (99.67% standalone, Table 37).

V Multi-Verifier NLI Results

Three-verifier BH correction. Table 27 applies Benjamini–Hochberg correction across all 15 Mistral-7B verifier–weight conditions (m=15, α=0.05; distinct from the m=12 cross-generator analysis in §6). Ten of 15 conditions remain significant after correction, confirming that multi-verifier NLI gains are robust to multiple testing.

Table 27: BH correction across 3 verifiers $\times$ 5 weights on Mistral-7B ProofWriter (n=600, m=15). ✓: $p_{\mathrm{BH}} < 0.05$.

Verifier	w=1	w=2	w=3	w=4	w=5
DeBERTa-MNLI		✓	✓	✓	✓
RoBERTa-MNLI			✓	✓	✓
BART-MNLI			✓	✓	✓

10/15 significant at α=0.05 after BH correction. All w=1 conditions and RoBERTa w=2 are non-significant.

W Capability Gap Data Points

Table 3 reports the controlled cross-generator comparison with DeBERTa-large-mnli on ProofWriter (n=600). We extend this analysis by pairing Mistral-7B with two additional NLI architectures: BART-large-mnli (46.17% standalone, combo +5.17 pp, $p = 0.0002$) and RoBERTa-large-mnli (37.67% standalone, combo +4.17 pp, $p = 0.003$). All three NLI architectures produce significant gains on Mistral-7B at $w \geq 3$.

Across all six ProofWriter verifier-generator pairs at w=3 (three verifiers × Mistral-7B plus three negative-gap generators × DeBERTa), combo gain decreases monotonically with the capability gap (Spearman $\rho = 1.0$, exact permutation $p = 0.003$, $n = 6$; however, four of six conditions share the DeBERTa verifier, which may reduce effective degrees of freedom—see Limitations; Figure 8). Significant w=3 gains are observed only for Mistral-7B (SC 39.33%, below the verifier); the three above-verifier generators show no significant improvement at any w. After BH correction (m=12), two of twelve conditions survive: Mistral-7B w=3 (adj. $p = 6.78 \times 10^{-5}$) and w=5 (adj. $p = 7.41 \times 10^{-6}$).

Robustness check. Including all nine conditions (three verifiers × three weight levels on Mistral-7B) yields Spearman $\rho = 0.836$ (exact permutation $p = 0.007$, $n = 9$), confirming that the monotonic trend is robust to the choice of weight parameter.

Table 28: Full cross-generator NLI combo results on ProofWriter depth-5 (n=600) with Benjamini–Hochberg correction (m=12, α=0.05). All conditions use DeBERTa-large-mnli verifier (55.00% standalone). Two of twelve conditions survive correction (Mistral-7B w=3 and w=5). *** $p_{\text{BH}} < 0.001$.

Generator	w	Combo (%)	Δ (pp)	p_{raw}	p_{BH}	
Mistral-7B (SC 39.33%, gap +15.67)	1	40.17	+0.83	.359	.721	
	3	46.00	+6.67	1.13×10^{-5}	6.78×10^{-5}	***
	5	50.33	+11.00	6.18×10^{-7}	7.41×10^{-6}	***
LLaMA-3.1-8B (SC 72.50%, gap −17.50)	1	73.33	+0.83	.359	.721	
	3	73.33	+0.83	.665	.886	
	5	72.00	−0.50	.871	1.000	
Qwen2.5-14B (SC 74.00%, gap −19.00)	1	74.17	+0.17	1.000	1.000	
	3	74.67	+0.67	.503	.755	
	5	75.17	+1.17	.360	.721	
Qwen3-14B (SC 85.83%, gap −30.83)	1	86.00	+0.17	1.000	1.000	
	3	86.33	+0.50	.453	.755	
	5	87.00	+1.17	.210	.721	

Full NLI combo with BH correction.

BART-large-mnli cross-generator results.

RoBERTa-large-mnli cross-generator results.

FOLIO NLI combo results.

Table 29: Cross-generator NLI combo with BART-large-mnli verifier (46.17% standalone) on ProofWriter depth-5 (n=600, w=3). Only Mistral-7B (below the verifier) gains significantly after BH correction.

Generator	SC (%)	Combo (%)	Δ (pp)	p_{BH}
Mistral-7B (gap +6.84)	39.33	44.50	+5.17	.003*
LLaMA-3.1-8B (gap −26.33)	72.50	72.50	±0.00	—
Qwen2.5-14B (gap −27.83)	74.00	75.50	+1.50	n.s.
Qwen3-14B (gap −39.66)	85.83	86.50	+0.67	n.s.

Table 30: Cross-generator NLI combo with RoBERTa-large-mnli verifier (37.67% standalone) on ProofWriter depth-5 (n=600, w=3). Only Mistral-7B (marginally above the verifier) gains significantly after BH correction.

Generator	SC (%)	Combo (%)	Δ (pp)	p_{BH}
Mistral-7B (gap −1.66)	39.33	43.50	+4.17	.016*
LLaMA-3.1-8B (gap −34.83)	72.50	71.33	−1.17	n.s.
Qwen2.5-14B (gap −36.33)	74.00	74.17	+0.17	n.s.
Qwen3-14B (gap −48.16)	85.83	86.50	+0.67	n.s.

LogiQA NLI combo results.

Domain-adapted verifier results.

Cross-model voting ablation.

Generator×Extractor 2×2 Matrix. Table 39 reports the full 2×2 generator×extractor matrix on FOLIO (n=204). The generator dominates accuracy (row difference ∼25 pp vs. column difference ∼3 pp), contrasting with ProofWriter where the extractor dominates (Δ range ±40–50 pp). Cross-extractor McNemar tests: Mistral-7B gen, $\Delta =$ +2.94 pp, $p = 0.143$ (n.s.); Qwen3-14B gen, $\Delta =$ +2.94 pp, $p = 0.021$ (statistically significant but practically small: 6/204 problems changed). The identical net Δ is coincidental—the two generators have different discordant-pair distributions that happen to yield the same net advantage. BR > 1 in all eight cells across both datasets (range 1.94–2.70), confirming SCB universality under cross-model extraction.

X Single-Trace and Aggregation Statistics

Extraction temperature sensitivity. Controlling for the trace set (excluding $T_{\text{ext}} = 0.3$, which uses a different seed), all conditions yield $\Delta \leq +1.47$ pp (n.s.), indicating that extraction temperature has no significant effect on SICA accuracy. This supports the structural nature of symbolic confirmation bias: the bottleneck is what gets extracted, not the temperature at which extraction occurs.

Multi-seed replication. On ProofWriter, all eight Mistral-7B and LLaMA-8B seeds yield nega-

Table 31: NLI combo on FOLIO (n=204) with DeBERTa-large-mnli verifier (49.51% standalone). No condition reaches significance; all generators exceed the verifier's standalone accuracy, consistent with the capability-gap pattern. SC baselines match the cross-model ablation traces (Table 38).

Generator	w	SC (%)	Combo (%)	Δ (pp)	p
	1		56.37	+0.49	n.s.
Mistral-7B	3	55.88	58.82	+2.94	n.s.
	5		59.80	+3.92	n.s.
	1		65.69	−0.49	n.s.
LLaMA-3.1-8B	3	66.18	65.20	−0.98	n.s.
	5		64.22	−1.96	n.s.
	1		76.96	+0.98	n.s.
Qwen2.5-14B	3	75.98	75.49	−0.49	n.s.
	5		73.53	−2.45	n.s.
	1		83.33	±0.00	n.s.
Qwen3-14B	3	83.33	84.80	+1.47	n.s.
	5		83.33	±0.00	n.s.

Table 32: NLI combo on FOLIO (n=204) with BART-large-mnli verifier (47.06% standalone). No positive improvement reaches significance; for Qwen2.5-14B the weak verifier produces a significant *degradation* at w=5 (p=.039), consistent with the capability-gap pattern.

Generator	w	SC (%)	Combo (%)	Δ (pp)	p
	1		54.90	−0.98	n.s.
Mistral-7B	3	55.88	56.37	+0.49	n.s.
	5		56.86	+0.98	n.s.
	1		67.65	+1.47	n.s.
LLaMA-3.1-8B	3	66.18	67.65	+1.47	n.s.
	5		64.71	−1.47	n.s.
	1		76.47	+0.49	n.s.
Qwen2.5-14B	3	75.98	75.00	−0.98	n.s.
	5		72.55	−3.43	.039*
	1		83.33	±0.00	n.s.
Qwen3-14B	3	83.33	84.31	+0.98	n.s.
	5		82.84	−0.49	n.s.

Table 33: NLI combo on FOLIO (n=204) with RoBERTa-large-mnli verifier (46.57% standalone). Cross-verifier consistency: no positive gain reaches significance, and Qwen2.5-14B shows a significant *degradation* at w=5 (p=.012)—the same capability-gap signature as BART.

Generator	w	SC (%)	Combo (%)	Δ (pp)	p
	1		56.37	+0.49	n.s.
Mistral-7B	3	55.88	57.84	+1.96	n.s.
	5		58.82	+2.94	n.s.
	1		65.69	−0.49	n.s.
LLaMA-3.1-8B	3	66.18	66.67	+0.49	n.s.
	5		64.22	−1.96	n.s.
	1		75.98	±0.00	n.s.
Qwen2.5-14B	3	75.98	75.00	−0.98	n.s.
	5		71.57	−4.41	.012*
	1		83.33	±0.00	n.s.
Qwen3-14B	3	83.33	84.31	+0.98	n.s.
	5		83.82	+0.49	n.s.

Table 34: NLI combo on LogiQA (n=200) with DeBERTa-large-mnli verifier (26.50% standalone, near the 25% random baseline for 4-way classification). No condition reaches significance.

Generator	w	SC (%)	Combo (%)	Δ (pp)	p
	1		54.00	+1.00	n.s.
Mistral-7B	3	53.00	54.00	+1.00	n.s.
	5		53.50	+0.50	n.s.
	1		58.50	−1.00	n.s.
LLaMA-3.1-8B	3	59.50	58.00	−1.50	n.s.
	5		58.00	−1.50	n.s.
	1		78.50	−0.50	n.s.
Qwen2.5-14B	3	79.00	78.00	−1.00	n.s.
	5		78.00	−1.00	n.s.
	1		81.00	±0.00	n.s.
Qwen3-14B	3	81.00	80.50	−0.50	n.s.
	5		80.50	−0.50	n.s.

tive Δ (range −1.33 to −7.50 pp), with all LLaMA-8B McNemar tests reaching $p < .001$. The consistency across seeds confirms that negative Δ on ProofWriter is not an artifact of any single random sample.

X.1 Sub-population Analysis

On FOLIO-204 with the corrected SC baseline (alphabetical tiebreaking), SICA improves over SC by +3.4 pp overall (55.9% → 59.3%), though the aggregate McNemar test is marginally non-significant (p=.092). The gain concentrates in the mid-constraint tertile (+9.4 pp, p=.031), where sufficient symbolic evidence exists for MaxSAT to meaningfully re-rank answers; in the low- and high-constraint groups the methods perform comparably. On ProofWriter-600, SICA shows a small uniform degradation consistent with the main results, attenuating in the high-constraint tertile (Δ=−0.5 pp, p=1.0). We note that with six subgroup tests, one significant result at α=0.05 is expected by chance; the mid-constraint finding should be interpreted as exploratory.

Individual intervention significance. Of the 27 training-free remediation strategies (Figure 4), the two largest positive effects are Cross-model Qwen3→Mistral (+0.98 pp) and Generational (+0.49 pp); neither approaches statistical significance ($p > 0.5$ for both). The largest negative effects are LLM-Debate (−18.6 pp, $p < 0.001$) and Contrastive Scoring (−9.3 pp, $p < 0.001$). After BH correction across all 27 comparisons, only these two negative results remain significant. No strategy produces a significant positive effect under any correction procedure.

Table 35: NLI combo on LogiQA (n=200) with BART-large-mnli verifier (33.00% standalone). Near-random NLI accuracy yields no significant combo improvements.

Generator	w	SC (%)	Combo (%)	Δ (pp)	p
Mistral-7B	1	53.00	53.50	+0.50	n.s.
	3		53.50	+0.50	n.s.
	5		52.50	−0.50	n.s.
LLaMA-3.1-8B	1	59.50	59.50	±0.00	n.s.
	3		59.50	±0.00	n.s.
	5		58.50	−1.00	n.s.
Qwen2.5-14B	1	79.00	78.50	−0.50	n.s.
	3		78.50	−0.50	n.s.
	5		77.50	−1.50	n.s.
Qwen3-14B	1	81.00	81.50	+0.50	n.s.
	3		81.00	±0.00	n.s.
	5		81.50	+0.50	n.s.

Table 36: NLI combo on LogiQA (n=200) with RoBERTa-large-mnli verifier (31.50% standalone). All three NLI verifiers produce the same null result on LogiQA.

Generator	w	SC (%)	Combo (%)	Δ (pp)	p
Mistral-7B	1	53.00	54.00	+1.00	n.s.
	3		53.50	+0.50	n.s.
	5		53.50	+0.50	n.s.
LLaMA-3.1-8B	1	59.50	59.00	−0.50	n.s.
	3		58.50	−1.00	n.s.
	5		58.50	−1.00	n.s.
Qwen2.5-14B	1	79.00	78.50	−0.50	n.s.
	3		78.50	−0.50	n.s.
	5		77.50	−1.50	n.s.
Qwen3-14B	1	81.00	81.00	±0.00	n.s.
	3		80.50	−0.50	n.s.
	5		81.00	±0.00	n.s.

Y Extended Discussion

SCB versus CoT unfaithfulness. SCB is distinct from CoT unfaithfulness (Turpin et al., 2023) and CoT-level confirmation bias (Wan et al., 2025; Jhaveri et al., 2026). CoT unfaithfulness concerns whether stated reasoning reflects actual computation; SCB concerns whether *self-extracted formal constraints* mirror the model's answer distribution. Even if traces were perfectly faithful, self-extraction would still inherit the distributional bias—the constraint $P(\text{answer\_A})$ reflects vote share, not logical validity. The shuffle control (§5.1) confirms this: randomizing constraint–answer associations preserves BR, showing the bias is in the trace-count distribution, not in extraction-specific errors. However, if traces are unfaithful, the mirage may be worse than our measurements suggest, because constraints would reflect not the model's reasoning but its post-hoc rationalization.

Table 37: Domain-adapted NLI combo on ProofWriter depth-5 (n=600). DeBERTa-large fine-tuned on ProofWriter OWA (99.67% standalone). All $p < 0.0001$. Complementarity: ratio of NLI-correct∩SC-incorrect to SC-correct∩NLI-incorrect.

Generator	w	Combo (%)	Δ (pp)	Compl. ratio
Mistral-7B SC 39.33%	1	42.17	+2.83	364:2
	3	52.00	+12.67	
	5	65.50	+26.17	
LLaMA-3.1-8B SC 72.50%	1	76.00	+3.50	164:1
	3	84.17	+11.67	
	5	91.67	+19.17	

Implications for neuro-symbolic practitioners. Our results do not imply that neuro-symbolic methods are inherently flawed—they imply that *self-extracted* constraints cannot provide independent verification. The key distinction is *when* formalization occurs relative to answer generation. Specification-generation approaches (e.g., SatLM (Ye et al., 2024), Logic-LM (Pan et al., 2023)) formalize the problem *before* generating candidate answers, avoiding the extraction bottleneck; SatLM's success confirms that the formal verification mechanism is sound when constraints are independently derived. Trace-extraction (this work) operates post-hoc on reasoning traces that already contain candidates, inheriting the distributional prior that generates SCB. When post-hoc verification is unavoidable, practitioners should employ external verifiers from a different architecture family (§6) and use the diagnostic framework in §7 to assess feasibility before deployment. The SCB diagnostic toolkit (below) provides detailed computation methods and threshold recommendations.

Test-time scaling implications. Vote diversity decreases with model capability: K_{eff} drops from 3.62 (Mistral-7B on ProofWriter) to 1.09 (Qwen3-14B thinking mode). This creates a fundamental tension for o1/R1-style scaling (Guo et al., 2025): longer reasoning chains produce more coherent but less diverse traces, reducing the marginal value of additional samples. Our Process-κ analysis (§5.1) reveals that this homogenization operates at the reasoning-process level, not just answer level—traces that disagree on final answers still share 69% of their reasoning structure. Cross-architecture ensembles offer a path forward, but the capability-gap condition (§6) implies that the verifier must be genuinely complementary, not just different.

SCB diagnostic toolkit. We recommend four zero-cost diagnostics, computable from sampled

Table 38: Cross-model voting ablation ($w \in \{3, 5\}$). "Cross LLM" replaces the NLI verifier with a second LLM's SC votes at weight w; "NLI" uses DeBERTa-large-mnli (55.00% standalone) at w=3. On ProofWriter, cross-model SC with a weak voter (Mistral-7B) degrades all strong generators—worsening at higher w—while NLI combo does not, indicating that structural independence—not merely model diversity—drives the escape.

Generator	Voter/Verifier	w	SC (%)	Combo (%)	Δ (pp)	p
ProofWriter (n=600)						
Mistral-7B	Qwen2.5-14B (cross)	3	39.33	47.67	+8.33	<.001
		5	39.33	55.33	+16.00	<.001
	DeBERTa NLI	3	39.33	46.00	+6.67	<.001
LLaMA-3.1-8B	Qwen2.5-14B (cross)	3	72.50	75.33	+2.83	.060
		5	72.50	75.33	+2.83	.125
	Mistral-7B (cross)	3	72.50	63.17	−9.33	<.001
		5	72.50	56.17	−16.33	<.001
	DeBERTa NLI	3	72.50	73.33	+0.83	.665
Qwen2.5-14B	LLaMA-8B (cross)	3	74.00	74.50	+0.50	.664
		5	74.00	76.00	+2.00	.096
	Mistral-7B (cross)	3	74.00	71.67	−2.33	.020
		5	74.00	69.67	−4.33	.002
	DeBERTa NLI	3	74.00	74.67	+0.67	.503
Qwen3-14B	Qwen2.5-14B (cross)	3	85.83	84.67	−1.17	.143
		5	85.83	83.33	−2.50	.006
	Mistral-7B (cross)	3	85.83	83.17	−2.67	.002
		5	85.83	81.17	−4.67	<.001
	DeBERTa NLI	3	85.83	86.33	+0.50	.453
FOLIO (n=204)						
Mistral-7B	Qwen2.5-14B (cross)	3	55.88	63.73	+7.84	.002
		5	55.88	68.63	+12.75	<.001
	DeBERTa NLI	3	55.88	58.82	+2.94	.210
LLaMA-3.1-8B	Qwen2.5-14B (cross)	3	66.18	70.10	+3.92	.134
		5	66.18	73.53	+7.35	.014
	DeBERTa NLI	3	66.18	65.20	−0.98	.824
Qwen2.5-14B	LLaMA-8B (cross)	3	75.98	76.96	+0.98	.727
		5	75.98	79.41	+3.43	.143
	DeBERTa NLI	3	75.98	75.49	−0.49	1.00
Qwen3-14B	Qwen2.5-14B (cross)	3	83.33	85.78	+2.45	.125
		5	83.33	86.76	+3.43	.039
	DeBERTa NLI	3	83.33	84.80	+1.47	.250

traces alone, to predict whether symbolic self-verification will help or hurt before deployment. *Bias Ratio* (BR; §5.1): for each problem, compute the fraction of constraints supporting the majority answer, divided by majority vote share. BR > 1 indicates constraints are biased toward the majority; BR $\gg 2$ (as in all nine models tested; Table 2) signals that extraction amplifies rather than checks the model's prior. Report as mean $\pm$ bootstrap 95% CI. *Effective-K* (K_{eff}; Eq. 1): compute Fleiss' κ across K traces, then $K_{\text{eff}} = 1 + (K-1)(1-\kappa)$. Threshold: $K_{\text{eff}} \leq 1.30$ never improves in our experiments—aggregation collapses to majority voting. $K_{\text{eff}} > 1.8$ is a necessary (not sufficient) condition for gain. *Degenerate Consensus* (C_0; §3.1): the fraction of problems where all K traces agree. $C_0 > 95\%$ renders aggregation moot; $C_0 < 5\%$ means the model is maximally uncertain and constraints tend to amplify noise. *ProcessSim* (Eq. 7): mean pairwise cosine similarity of TF-IDF trace vectors. ProcessSim > 0.6 indicates homogeneous reasoning; combined with BR > 2, this

Table 39: FOLIO 2×2 generator×extractor matrix (n=204). SC: Self-Consistency baseline. κ: answer-level Fleiss' kappa. BR: bias ratio.

Generator	Extractor	SC (%)	SICA (%)	Δ (pp)	BR
Mistral-7B	Mistral	55.88	56.86	+0.98	2.70
κ=0.46	Qwen3	55.88	59.80	+3.92	1.94
Qwen3-14B	Qwen3	83.33	84.80	+1.47	2.35
κ=0.79	Mistral	83.33	81.86	−1.47	2.66

Table 40: Single-trace accuracy (mean $\pm$ std over K=12 traces) and SC@12 majority vote on FOLIO (n=204). Aggregation gain = SC@12 − single-trace mean. SC@12 uses alphabetical tiebreaking (consistent with Table 1).

Model	Single (%)	Std	SC@12 (%)	Gain (pp)
Mistral-7B	46.41	2.80	55.88	+9.47
Qwen3-14B	79.37	1.98	84.80	+5.43
Qwen3-14B (think)	85.42	0.99	86.27	+0.86

Inter-trace standard deviation shrinks 2.8× from Mistral-7B to Qwen3-14B thinking mode, reflecting the diversity compression that underlies the aggregation ceiling (§6).

predicts SCB will dominate. A practitioner should compute all four on a held-out sample ($n \geq 50$). If $K_{\text{eff}} \leq 1.30$ or $C_0 > 95\%$ or (BR > 2 and ProcessSim > 0.6), symbolic self-verification is unlikely to help.

BR permutation null model. The permutation test referenced in §5.1 proceeds as follows:

1. For each problem p with extracted constraints $\{c_1, \ldots, c_M\}$ and source answers $\{a_1, \ldots, a_M\}$:
2. Randomly permute constraint-to-answer assignments within p (preserving per-trace constraint counts).
3. Compute BR_{null} from the permuted assignments.
4. Repeat 10,000 times to obtain the null distribution.
5. Compute p-value $= |\{r : \text{BR}^{(r)}_{\text{null}} \geq \text{BR}_{\text{obs}}\}|/10{,}000$.

Across all tested conditions, $\text{BR}_{\text{null}} \approx \text{BR}_{\text{obs}}$ ($p > 0.7$), confirming that the observed bias ratio is fully explained by trace-count geometry under source-attribution scoring (Observation 1), not by an independent extraction bias.

Future directions: trained extractors and beyond. Our oracle controls reveal that perfect extraction would yield +29.90 pp over SC on Mistral-7B FOLIO—self-extraction captures only ∼12% of this ceiling. This gap motivates several future directions. *Trained constraint extractors*: fine-tuning a small model (e.g., encoder-only) on human-annotated FOL from logic benchmarks

Table 41: Effect of extraction temperature T_{ext} on SICA accuracy (Mistral-7B, FOLIO n=204). † Uses a different trace set (SC = 54.41%); all other conditions share the same traces (SC = 55.88%).

T_{ext}	SICA (%)	Δ vs. SC (pp)	Avg. #Constr.
0.1	57.35	+1.47	69.3
0.3†	59.31	+3.43	68.8
0.5	56.37	+0.49	68.2
0.7	53.43	−2.45	67.3
1.0	55.39	−0.49	65.2

could produce genuinely discriminative constraints that break the distributional prior. The key design constraint is structural independence—the extractor should not share training data or architecture with the generator, mirroring the cross-architecture NLI escape (§6). *Multi-model ensembles*: rather than single cross-architecture pairs, combining M structurally independent verifiers could further reduce correlated errors. The capability-gap condition (§6) predicts that each verifier must exceed the generator on the target task to contribute positively; empirical verification across heterogeneous verifier pools is needed. *External knowledge grounding*: connecting to knowledge bases (Wikidata, domain ontologies) could inject verification signal absent from any model's parameters. This avoids SCB entirely by grounding constraints in external facts rather than self-generated traces. *SCB-aware aggregation*: current MAX-SAT aggregation treats all constraints equally. Weighting constraints by estimated independence (e.g., inverse ProcessSim with other traces' constraints) or by discriminative power (e.g., cross-validation on held-out traces) could partially mitigate SCB without requiring external verifiers.

SCB as an evaluation framework. The SCB metrics—BR, K_{eff}, ProcessSim, and CCR—can serve as a general evaluation framework for self-improvement methods beyond SICA. Any method that extracts structured representations from a model's own outputs (self-verification (Huang et al., 2024), self-refinement (Madaan et al., 2023), chain-of-verification (Dhuliawala et al., 2023), self-consistency (Wang et al., 2023)) faces the distributional prior that generates SCB. We propose that future work on self-improvement methods report: (1) BR on error cases (does the method's verification signal correlate with the model's majority answer, or does it provide independent information?); (2) K_{eff} or an analogous diversity metric (does the method generate genuinely diverse candidates, or does it amplify a single mode?); (3) CCR (does

Table 42: Multi-seed replication results across four model×dataset conditions. Each row uses an independent seed with fresh traces (K=12, T=0.7). Mistral-7B×FOLIO shows directionally unstable Δ (+0.16 ± 2.70 pp); all ProofWriter conditions show consistently negative Δ across all seeds ($p < .001$ for LLaMA-8B and Mistral-7B). Qwen3-14B results pending.

Seed	SC (%)	SICA (%)	Δ (pp)
Mistral-7B × FOLIO (n=204)[a]			
123	54.41	57.35	+2.94
456	55.88	55.88	0.00
789	59.31	56.86	−2.45
Mean ± SD	56.53 ± 2.51	56.70 ± 0.75	+0.16 ± 2.70
Qwen2.5-14B × FOLIO (n=204)			
42	77.50	75.49	−2.01
123	74.51	73.04	−1.47
456	74.02	74.02	0.00
789	74.02	72.55	−1.47
Mean ± SD	75.01 ± 1.67	73.78 ± 1.30	−1.24 ± 0.86
Mistral-7B × ProofWriter (n=600)			
42	39.33	36.50	−2.83
123	40.83	37.50	−3.33
456	39.83	38.50	−1.33
789	38.83	37.17	−1.67
Mean ± SD	39.70 ± 0.85	37.42 ± 0.83	−2.29 ± 0.95
LLaMA-3.1-8B × ProofWriter (n=600)			
42	72.50	65.00	−7.50
123	71.33	64.00	−7.33
456	71.67	66.17	−5.50
789	71.17	66.50	−4.67
Mean ± SD	71.67 ± 0.59	65.42 ± 1.14	−6.25 ± 1.34

[a]Original baseline (exp033) used no fixed seed and is excluded from seed statistics.

the method preferentially correct errors, or does it preferentially confirm them?). A method with BR < 1, high K_{eff}, and CCR > 0 demonstrates genuine self-improvement; BR > 1 with CCR < 0 indicates that the method suffers from SCB regardless of aggregate accuracy gains, which may reflect favorable base rates rather than corrective signal.

Additional scope considerations. Beyond the limitations discussed in the main text, several additional constraints bound our conclusions. Seven of our eleven model-domain conditions have $n \leq 60$, providing insufficient statistical power for confirmatory claims; power analysis shows that the minimum detectable effect exceeds 5 pp for these conditions (Table 6), so they serve as regime characterization rather than definitive tests. Our extraction prompts use a single zero-shot template per domain; alternative prompting strategies (few-shot exemplars, chain-of-thought extraction, or iterative refinement) might yield higher-quality constraints, though the oracle gap suggests the bottleneck is structural rather than prompt-specific. The constraint quality analysis (91.8% discrimi-

Table 43: SICA vs. SC accuracy by gold-label type (Mistral-7B). Corrected SC baseline uses alphabetical tiebreaking. Δ pp = SICA $-$ SC. p-values from exact McNemar test (two-sided).

Dataset	Label	n	SC	SICA	Δ pp	p
FOLIO-204	True	72	56.9	62.5	$+5.6$	.125
	False	63	42.9	42.9	± 0.0	1.0
	Unknown	69	66.7	71.0	$+4.3$	.250
	Overall	204	55.9	59.3	$+3.4$	.092
PW-600	True	200	8.5	6.0	-2.5	.063
	False	200	19.5	14.5	-5.0	.076
	Unknown	200	90.0	89.0	-1.0	.774
	Overall	600	39.3	36.5	-2.8	.014*

Table 44: SICA vs. SC accuracy by extracted constraint count tertile (Mistral-7B). Corrected SC baseline (alphabetical tiebreaking). Constraint counts are unique deduplicated constraints per problem.

Dataset	Tertile	n	$\bar{c}$	SC	SICA	Δ pp	p
FOLIO-204	Low [0–32]	74	21.7	62.2	63.5	$+1.4$	1.0
	Mid [33–47]	64	39.7	50.0	59.4	$+9.4$	.031*
	High [48–83]	66	58.7	54.5	54.5	± 0.0	1.0
PW-600	Low [0–39]	212	28.8	43.9	39.6	-4.3	.078
	Mid [40–52]	196	45.8	40.8	37.2	-3.6	.143
	High [53–99]	192	63.7	32.8	32.3	-0.5	1.0

native rate, 39.2% source-trace minority weight) covers Qwen2.5-14B on FOLIO + ProofWriter ($N = 804$); other model-domain combinations may exhibit different characteristics. Constraint extraction uses $T_{\text{ext}} = 0.3$ while trace sampling uses $T = 0.7$; alternative extraction temperatures might yield different results. Although we prompt for first-order logic formalizations, 99.0% of extracted constraints are propositional at the Z3 level, with only 1.0% using true quantifiers ($\forall$/$\exists$); whether models capable of genuine quantified reasoning would yield higher discriminative rates is untested. Deduplication uses syntactic equivalence (global average dedup ratio: 46.0%), so semantically equivalent constraints with different surface forms receive separate weights; Z3-based semantic deduplication removes an additional $\sim$10% of constraints and yields a small but significant improvement over no deduplication ($p = 0.013$), suggesting room for better dedup strategies. The GSM8K pilot ($n = 30$) illustrates the constraint-absent regime and should not be interpreted as confirmatory evidence.

Z Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.